DESIGN STUDY · 2026 EDITION

Tchoua Association Resource Planning
Complete design study

Designing a complete association management product: organisation, governance, tontines, contributions, savings, loans, solidarity, activities and resources.

168 pages | 36 domains | 144 sheets | 23,328 combinations

01 · ORIENTATIONS FOR PRODUCT LEADERSHIP

Summary and structural decisions

According to the manual provided, Tchoua has the essential objects of an association management package: association, member, activity, session and movement. Moving to ARP means making the rules explicit, the commitments calculable and the evidence accessible, then extending coverage to projects, resources, purchases and community services.

Decision	Consequence for the product
Common management core	Separate user account, membership, contract, debt, payment and allocation. Each module consumes the same references.
Separate funds and accounts	Where funds are held does not define their permitted use. Budget, reservation and liquidity become three separate measures.
Versioned rules	Store base, rate, unit, effective date, exceptions and approval decision. Replay historical calculations.
Non-deactivatable invariants	Isolation of associations, uniqueness of payments, audit, preservation of amounts and concurrency control.
Progressive extension	Stabilise the financial core before automating auctions, integrated payments and consolidation.
Risk-based test recipe	Cover normal cases, limits, errors, concurrency, recovery after failure and interactions between modules.

What must be decided before development

The product committee validates the priority models, the launch countries, the decision authorities, the treatment of debts on exit, the rounding policy and the funds authorised for each product. The competent experts validate the legal, accounting and payment frameworks required to activate the products concerned.

> **Scope of completeness.** A finished study cannot enumerate every future status or every possible amount. The completeness sought is structural: a catalogue of dimensions, constraints, combination models and test criteria. New exceptions are added as versioned rules, not as hidden processing.

02 · REFERENCES AND TRADE-OFFS TO PRESERVE

Critical reading of the documentation

The manual is the source of the declared existing system. It has not been superseded by the instructions it contains, nor has it been treated as proof that the code executes. The pages cited below are the physical page numbers of the PDF provided. The proposals in this study describe a target to be built and validated.

Point to clarify	Documentary finding	Design decision
Tontine automation	Calculations described pp. 31–35; automatic workflow flagged for completion p. 82.	Distinguish operational calculation from target automatic selection.
Multi-currency	Promise p. 3; currency per association p. 28; conversions to come p. 71.	Functional currency per entity and explicit conversion.
Visibility of histories	Global member view pp. 16 and 58; personal details restricted p. 13.	Aggregated transparency with rights over details and supporting documents.
Auction base	5% of the contribution rate, p. 32 and 76.	Do not conflate with the pot; have the base explicitly named.
Age for child aid	Under ten and over ten, p. 46–47.	Explicitly configure the ten-year case.
Successive sanctions	Scale of 1 000, 3 000 and 5 000, p. 37.	Choose addition, replacement or maximum.
HORIZON dividends	Interest 5 219 831; base 5 433 499, p. 44 and 54.	Justify the 213 668 difference; do not invent a product.
GA convening	30-day notice p. 51; FEV parameter 45 days p. 75.	The validated specific contract/status prevails.

Traceability of the need

Each domain includes a reference to the manual, identified rules and tests. The reference to the manual attests the origin of the topic; it does not mean that all the extensions in the sheet are already implemented. The proposed API contracts and data models must be compared against the repository before implementation.

03 · TRANSFORMATION PRIORITIES

Declared existing system and target scope

Domain	State described by the manual	Target of the study
Associations and members	Creation, sections, roles, documents and memberships described as operational.	Term dates, families, settlements and controlled inheritance.
Tontines	Shares, cycles, balances and parameters; automation still partial.	Atomic allocations, verifiable draws, complete auctions, defaults.
Contributions and funds	Calls, data entry, history and exports.	Debt / payment / allocation, commitments, budgets and balanced journal.
Loans and interest	Applications, repayments, interest and dividends.	Catalogue of methods, total cost, reserved guarantees, restructuring.
Solidarity	Catalogue, applications and aid.	Committed caps, cumulative totals, third-party payers, recourse, fund forecasts.
Operator payments	Configurable means, integrations to be completed.	Authenticated connectors, idempotency, reconciliation and refunds.
Consolidation	Hierarchy available, national aggregation to come.	Dated scopes, reciprocal eliminations, traceable conversions.
Accounting	Management reports; full chart of accounts not integrated.	Structured journal, then adaptation to the framework validated by a specialist.
Resources and community	Activity types and navigation entries mentioned.	Projects, purchases, stocks, assets, training and distinctions detailed.
Mobile and operations	Responsive, backups and monitoring described.	Controlled synchronisation, tested recovery, migration and observability.

Qualification criterion. A function is “documented” if the manual describes it. It is “validated” only after test evidence on the target environment. The backlog must keep these two statuses separate.

04 · MODEL LIBRARY

Association profiles and starting configurations

Models speed up configuration without locking an association into a single activity. They copy a version of parameters, then require validation of local choices. Optional modules remain independent, subject to their financial and security dependencies.

Profile	Base to activate	Variants and watch points
Neighbourhood tontine	Memberships, rotation, contributions, fund, meetings.	Fractions, absence, default after payout, exit and surplus.
Family mutual aid group	Households, aid, contributions, events.	Family ties, caps per person or household, beneficiaries.
SLD-type section	Auctions, solidarity, investment, refreshments.	Separate funds, national inheritance and auction-base ambiguities to resolve.
FEV-type federation	Sections, delegates, national fund, aid, consolidation.	Unique persons versus memberships, transfers back and limited access.
HORIZON-type investment	Capital contributions, loans, repayments, interest, dividends.	Capital as at a date, justified results, liquidity and related parties.
Agricultural grouping	Nature, stocks, purchases, marketing, projects.	Units, quality, losses and absence of liquidity before sale.
Cultural or sports club	Memberships, events, equipment, volunteering.	Capacity, equipment, represented minors and insurance to be validated.
Development association	Donations, grants, projects and purchases.	Earmarked funds, proof of expenditure and donor reports.
Professional network	Private directory, services, training, governance.	Confidentiality, conflicts of interest and quality of services.
Multi-country diaspora	Sections, contributions, foreign exchange, projects, solidarity.	Currencies, providers, country framework and transfer costs.

Cooperative and mutual are proposed as functional profiles to be adapted. Their qualification, their obligations and the products they can carry cannot be inferred from activating a software model.

05 · COMMON PRODUCT ARCHITECTURE

Functional map and dependencies

USER EXPERIENCES

Member · board · treasurer · controller · federation · support

ORGANISATION AND CONTRACTS

Identities · memberships · bodies · terms of office · rules · documents

PRODUCTS AND RESOURCES

Tontines · savings · loans · aid · projects · stocks · events

COMMON FINANCIAL CORE

Obligations · payments · allocations · reservations · journal

PLATFORM SERVICES

Authorisations · audit · files · notifications · connectors · exports

Dependencies that determine activation

Desired activation	Mandatory prerequisites
Tontine distribution	Contract, subscriptions, remaining rights, confirmed funds and concurrency checks.
Loan with guarantee	Product, cap, liquidity, consent and guarantee reservation.
Social aid	Catalogue, event, rights, caps, budget and payment circuit.
Consolidation	Dated scope, common reference data, currencies and identifiable reciprocal flows.

Deactivation hides new subscriptions but preserves the consultations, corrections and settlements required by existing obligations. The security and audit core remains active for all modules.

06 · NAVIGATING THE 144 OPERATIONAL SHEETS

The 36 design domains

Each domain has four sheets: A · scope, options and invariants; B · detailed form and interface rules; C · workflow, exceptions and transactional control; D · combinations, test scenarios and indicators.

01 Identity and forms of organization
A · B · C · D

02 Federations, sections and inheritance
A · B · C · D

03 Memberships, members and exits
A · B · C · D

04 Families, beneficiaries and confidentiality
A · B · C · D

05 Governance, mandates and votes
A · B · C · D

06 Meetings, calendar and attendance
A · B · C · D

07 Discipline, sanctions and appeals
A · B · C · D

08 Documents, internal rules and signatures
A · B · C · D

09 Rule engine and activity contracts
A · B · C · D

10 Contributions, calls and allocations
A · B · C · D

11 Payments received and reconciliation
A · B · C · D

12 Cash accounts, budgets and entries
A · B · C · D

13 Tontine share subscriptions and fractions
A · B · C · D

14 Allocations, rotations and draws
A · B · C · D

15 Auctions, discounts and sharing of the proceeds
A · B · C · D

16 Tontine cycles, closings and defaults
A · B · C · D

17 Savings, deposits, withdrawals and interest
A · B · C · D

18 Loans, applications, caps and decision
A · B · C · D

19 Interest schedules and cost of credit
A · B · C · D

20 Repayments, arrears and restructurings
A · B · C · D

21 Security, guarantees and commitments
A · B · C · D

22 Solidarity catalogue and social entitlements
A · B · C · D

23 **Aid applications, payments and appeals**
A · B · C · D

24 **Solidarity reserves and liquidity forecasting**
A · B · C · D

25 **Capital investment and dividends**
A · B · C · D

26 **Projects, grants, donations and campaigns**
A · B · C · D

27 **Group purchasing, inventory and in-kind contributions**
A · B · C · D

28 **Physical assets, volunteering and time**
A · B · C · D

29 **Events, refreshments and logistics**
A · B · C · D

30 **Community, services, training and awards**
A · B · C · D

31 **Reports, dashboards and consolidation**
A · B · C · D

32 **Security, authorisations and data protection**
A · B · C · D

33 **Data architecture, API and integrations**
A · B · C · D

34 **Mobile, accessibility, languages and notifications**
A · B · C · D

35 **Administration, migration, operations and support**
A · B · C · D

36 **Internationalisation, currencies and country models**
A · B · C · D

Design appendices: formulas and worked cases; compatibility and combinatorial testing; integrated journeys; data and transactions; trajectory and acceptance; glossary and decisions.

D01

DOMAIN 01 · OPERATIONAL SHEET

Identity and forms of organization

The framework distinguishes the form of organization, the declared legal status and the operating profile. A friendly society can run a tontine without becoming a tontine in the sense of its identity. A federation can bring together autonomous entities or internal sections; this difference governs ownership of funds, authorizations and consolidation. Configuration must support the family, the neighbourhood, the village, the professional network, the cultural or religious grouping, the sports club, the diaspora, the mutual society and the cooperative. These categories are functional models and do not automatically confer a legal qualification.

Manual reference: p. 18–23, 65–66. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Structure	Standalone association, internal section, federation, project collective	Determines the scope of decisions and accounts.
Purpose	Mutual aid, savings, investment, culture, sport, development	Several purposes may coexist with separate budgets.
Membership	Individual, family, legal person, mixed	A representative must not be confused with the entity represented.
Territory	Local, regional, national, diaspora	Keep country, time zone and functional currency per entity.
Term	Unlimited, fixed-term project, renewable cycle	Reaching the term opens a controlled liquidation.
Template	Blank, FEV, SLD, HORIZON, custom template	Copy an identified version; do not impose updates.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behaviour
ORG01	A stable identifier survives a change of name; the former name remains in the history.
ORG02	Activation requires recorded officers, minimum rules, currency and creation decision.
ORG03	A dissolution blocks new commitments but authorizes approved liquidation operations.
ORG04	The template is neither a filed legal status nor an authorization to carry on a regulated activity.

B · Form and data-entry experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced by fictitious values. Final validation runs server-side.

Functional field	Type or content	Validation and input assistance
D01-F01 · Name	Text, 160 characters	Mandatory; check for near-duplicates without blocking legitimate namesakes.
D01-F02 · Functional form	Extensible list	Choose a template; a change does not alter vested rights.
D01-F03 · Declared status	Text and document	Distinguish declared, verified and not provided; keep the evidence.
D01-F04 · Country and territory	Codes and text	Country required; region and locality optional depending on context.
D01-F05 · Creation date	Local date	Cannot be later than activity already imported without justification.
D01-F06 · Functional currency	Currency code	Lock after the first write; conversion through a dedicated procedure.
D01-F07 · Initial officer	Member reference	Named invitation and acceptance before activation.
D01-F08 · Starting template	Identifier and version	Show the copied modules and the parameters still to be completed.

COMMON CONTEXT AND EDITING RIGHTS

The screen always shows the association, activity or file concerned, the currency where necessary, the status and the version. The author, creation date and identifier are generated by the system. The role authorised at the workflow step determines the available actions; a hidden button does not replace the access control.

> **Summary before validation.** The screen presents the inputs, the expected result and the errors before the final action. After submission, any significant correction displays the differences; errors target the field concerned and preserve the other entered data.

Screen actions: save a draft, verify, submit, request additional information, approve or reject according to role.

C · Workflow, exceptions and transactional control

Actor	Action	Control
Founder	Create the file	Unique identity and template made explicit
Secretary	Attach the governing texts	Legible, versioned documents
Board	Configure powers	Separate officers and dated mandates
Controller	Review the file	Contradictions flagged
Statutory authority	Activate	Decision and effective date retained
System	Open the services	Log and invitations generated

EXCEPTIONS AND RECOVERY

Trigger	Response	Expected trace or recovery
Namesake	Present the matches	Create a separate identifier after verification.
Abandoned creation	Archive the draft	Generate neither debt nor fund call.
Change of form	Prepare an amendment	Maintain historical commitments.
Dissolution with loans	Move to liquidation	Keep recoveries and third-party accounts.

Trigger	Response	Expected trace or recovery
Officer unavailable	Activate a substitute	Require a dated decision, never a shared account.

D · Combinations, test recipes and indicators

COMBINATIONS AND QUALIFICATIONS

Combination of functions	Qualification	Validity condition
Friendly society + tontine	Compatible	A separate activity carries the tontine contract.
Cooperative + in-kind	Conditional	Units, stock and valuation must be configured.
Federation + sections	Compatible	Define autonomy and inheritance rules.
Dormant association + new loan	Prohibited by default	Reactivate before any new commitment.
Temporary project + solidarity	Conditional	Plan what becomes of rights after the project ends.

TEST SCENARIOS

Case	Data or trigger	Verifiable result
D01-T01 · Creating a friendly society	Identity, currency and board provided	Activation creates an isolated space and no implicit contribution.
D01-T02 · SLD template copied	Local amount changed	The source template and the other associations remain unchanged.
D01-T03 · Incomplete file	Officer missing	Activation is refused with the list of missing fields.
D01-T04 · Dissolution	Loan still open	The loan remains viewable and recoverable in liquidation.
D01-T05 · Renaming	Old invoices present	Old documents keep the name under which they were issued.

MANAGEMENT INDICATORS

Indicator	Formula	Reading
File completeness	Validated fields and evidence / required items	Measure before activation, without counting optional fields.
Active associations	Active entities at the reference date	Exclude drafts and duplicate imports.

D02

DOMAIN 02 · OPERATIONAL SHEET

Federations, sections and inheritance

The hierarchy must carry explicit relationships: administrative attachment, affiliation and accounting ownership are three distinct notions. The target allows several levels but forbids cycles. A section applies a locked national baseline and may override the parameters that the parent declares modifiable. The federation does not automatically receive named access to all local files. Aggregated statistics, federal contributions and transfers are handled by separate contracts. National consolidation is an extension flagged in the manual as still to be built.

Manual reference: p. 13, 21–23, 75, 82. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Relationship	Internal section, institutional member, partner	Defines read rights and the handling of transfers.
Depth	National, region, department, local	Limit by policy without hard-coding the number of levels.
Inheritance	Mandatory, default value, free	Show the origin and effective value of each rule.
Federal contribution	Fixed, per member, percentage, brackets	Produce a reciprocal receivable and a verifiable breakdown.
Delegates	Fixed, proportional, cap	Specify rounding, reference date and eligible members.
Consolidation	Aggregated management, accounting scope	Eliminate internal flows only within the retained scope.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behaviour
FED01	No parent can be its own descendant, including during a concurrent change.
FED02	A blocking national rule prevails over a local override; contracts already concluded keep their version.
FED03	A transfer changes two reciprocal positions without creating fictitious consolidated income.
FED04	A change of parent takes effect on a date; it does not move entries retroactively.

B · Form and data-entry experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced by fictitious values. Final validation runs server-side.

Functional field	Type or content	Validation and input assistance
D02-F01 · Child entity	Organization reference	Require an existing entity and the consent of its representatives.
D02-F02 · Parent entity	Organization reference	Search within the authorized affiliations; check that no cycle exists.
D02-F03 · Nature of the link	Enumeration	Section or affiliation; do not infer this choice from the name.
D02-F04 · Effective date	Date	Close the former link without overlapping incompatible periods.
D02-F05 · Inherited baseline	Policy version	Show the differences and the locked parameters.
D02-F06 · Remittance	Formula and frequency	Specify gross or net basis and the treatment of exemptions.
D02-F07 · Delegates	Rate, rounding, cap	FEV example to be specified: 10% and cap 11; rounding to be decided.
D02-F08 · Federal visibility	Aggregated or authorized named access	Restrict sensitive documents to authorized holders.

COMMON CONTEXT AND EDITING RIGHTS

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> **Summary before validation.** The screen presents the inputs, the expected result and the errors before the final action. After submission, any significant correction displays the differences; errors target the field concerned and preserve the other entered data.

Screen actions: save a draft, verify, submit, request additional information, approve or reject according to role.

C · Workflow, exceptions and transactional control

Actor	Action	Control
Section	Request attachment	Identity and mandate verified
Federation	Review the request	Compatibility of the rules
System	Compare the rules	Blocking conflicts displayed
Both boards	Approve the link	Two decisions retained
System	Apply on the effective date	Versions and rights updated
Controller	Reconcile the flows	Reciprocal balances in agreement

EXCEPTIONS AND RECOVERY

Trigger	Response	Expected trace or recovery
Hierarchical cycle	Reject the attachment	Keep the path that caused the refusal.
Contrary local rule	Block publication	Offer the national value and a remedy.
Section departure	Close the commitments	Federal receivables survive the departure.
Duplicate member count	Deduplicate persons	Also keep the distinct number of memberships.
Remittance not received	Create a reciprocal discrepancy	Do not mark the transfer as reconciled.

D · Combinations, test recipes and indicators

COMBINATIONS AND QUALIFICATIONS

Combination of functions	Qualification	Validity condition
Federation + medical data	Restricted	Transmit aggregates, not supporting documents.
Sections + different currency	Conditional	Convert under a documented policy.
Inheritance + loan amendment	Conditional	The new rule does not rewrite the existing contract.
Multiple affiliation + vote	Conditional	Define the representation mandate in each body.
Transfer + consolidation	Compatible	Keep a common elimination identifier.

TEST SCENARIOS

Case	Data or trigger	Verifiable result
D02-T01 · Three levels	National, region, section	The national rate applies unless an authorized override exists.
D02-T02 · Circular attachment	A parent of B and the reverse request	The transaction is rejected with no partial modification.
D02-T03 · Transfer 100 000	Section to federation	The consolidated cash total does not change.
D02-T04 · Multiple membership	One person in two sections	Headcount of persons equals one; memberships equal two.
D02-T05 · Dated departure	Exit on 30 June	May reports keep the former scope.

MANAGEMENT INDICATORS

Indicator	Formula	Reading
Inter-entity discrepancy	Sum of receivables minus reciprocal payables	Present discrepancies by counterparty and date.
Baseline compliance	Compliant sections / sections checked	Separate authorized derogations from anomalies.

D03

DOMAIN 03 · OPERATIONAL SHEET

Memberships, members and exits

Membership is a contract between a person and an association, distinct from the login account. A member without a smartphone can be registered and assisted by an authorized agent; that agent signs for their own actions. The file follows application, review, acceptance, activation, suspension and exit. Voting, aid, credit and tontine rights depend on separate conditions. A removal does not settle a debt, and a death does not automatically transfer funds to the first person who comes forward. Legal persons have representatives whose mandates are dated.

Manual reference: p. 14–17, 18, 77, 81. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Admission	Open, invitation, sponsorship, committee	Each route leads to a decision and evidence.
Member class	Active, honorary, benefactor, junior, institutional	Financial and political rights are configured separately.
Entrance fee	Fixed, by category, waiver	Distinguish a vested fee from a refundable deposit.
Seniority	Admission, activation, first contribution date	Show the date used for each benefit.
Suspension	Administrative, disciplinary, arrears	Specify frozen rights and the obligations that continue.
Exit	Resignation, removal, death, transfer	Open an individual settlement and a final statement.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behaviour
MEM01	The member number is unique within the association without being the person's global identifier.
MEM02	Accepting an application and paying the entrance fee are distinct facts.
MEM03	Suspension of rights destroys neither savings, nor debt, nor proof of payment.
MEM04	Every exit produces a contradictory statement of receivables, guarantees and unserved rights.

B · Form and data-entry experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced by fictitious values. Final validation runs server-side.

Functional field	Type or content	Validation and input assistance
D03-F01 · Member identity	Person reference	Duplicate detection by assisted matching, without automatic merging.
D03-F02 · Category	Configured list	Show the associated rights and conditions before submission.
D03-F03 · Application date	System timestamp	Not editable by the applicant.
D03-F04 · Sponsors	List of members	Minimum number, seniority and no self-sponsorship.
D03-F05 · Admission documents	Typed documents	Collect only the documents required by the template.
D03-F06 · Entrance fee	Amount and waiver	A waiver requires a reason and an authority; keep the original debt.
D03-F07 · Consents	Separate choices	Separate membership, communications and public directory.
D03-F08 · Exit reason	Code and comment	Require effective date, controller and destination of remaining rights.

COMMON CONTEXT AND EDITING RIGHTS

The screen always shows the association, activity or file concerned, the currency where necessary, the status and the version. The author, creation date and identifier are generated by the system. The role authorised at the workflow step determines the available actions; a hidden button does not replace the access control.

> **Summary before validation.** The screen presents the inputs, the expected result and the errors before the final action. After submission, any significant correction displays the differences; errors target the field concerned and preserve the other entered data.

Screen actions: save a draft, verify, submit, request additional information, approve or reject according to role.

C · Workflow, exceptions and transactional control

Actor	Action	Control
Applicant	Submit the file	Required fields and documents
Secretary	Verify identity	Duplicates and sponsors
Committee	Decide on admission	Reason and quorum
Treasurer	Record the fee	Receipt or waiver
System	Activate the rights	Date and category frozen
Board	Close on exit	Contradictory statement

EXCEPTIONS AND RECOVERY

Trigger	Response	Expected trace or recovery
Payment before admission	Hold in suspense	Allocate after admission or refund on decision.
Application refused	Notify the reason	Limit retention of documents according to policy.
Duplicate file	Propose a match	Merge authorized with an identifier log.
Exit after receiving the pot	Maintain the future debt	Schedule recovery and guarantees.

Trigger	Response	Expected trace or recovery
Death with savings	Freeze the final handover	Verify the beneficiaries and trace the decision.

D · Combinations, test recipes and indicators

COMBINATIONS AND QUALIFICATIONS

Combination of functions	Qualification	Validity condition
Honorary member + vote	Conditional	Explicit right, never inferred from the waiver.
Suspension + emergency aid	Conditional	Derogation circuit with a reasoned decision.
Exit + guarantee	Conditional	Release waits for a replacement or the end of the risk.
Minor + savings	Conditional	Representation and withdrawals subject to the approved framework.
Legal person + several representatives	Compatible	A single economic right remains held by the entity.

TEST SCENARIOS

Case	Data or trigger	Verifiable result
D03-T01 · Free membership	Waiver approved	The member is active without a false cash receipt entry.
D03-T02 · Insufficient sponsors	One sponsor instead of two	The file remains incomplete, with no activation.
D03-T03 · Two associations	Same person, different categories	Rights remain specific to each membership.
D03-T04 · Removal with loan	Outstanding principal positive	The loan remains in the outstanding balance and authorized reminders.

Case	Data or trigger	Verifiable result
D03-T05 · Re-admission	Former member number archived	Create a new period without duplicating the person.

MANAGEMENT INDICATORS

Indicator	Formula	Reading
Admission lead time	Activation minus complete submission	Exclude waiting periods attributable to missing documents.
Retention rate	Members still active / initial cohort	Publish the period and the definition of exits.

D04

DOMAIN 04 · OPERATIONAL SHEET

Families, beneficiaries and confidentiality

Family aid requires a reference of related persons, events and evidence. The family link must not be a simple text copied into each request: its validity period, its verification status and the rights concerned must be known. The SLD rules on declared spouse, first marriage or birth are local examples from the document, not universal constraints of the package. The target makes them configurable and subject to validation. Health data is isolated from the information needed for accounting and transparency reports.

Manual reference: p. 46–48, 58–59. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Beneficiary unit	Person, household, member, dependent	Specify which unit carries the cap.
Link	Spouse, child, parent, dependent	Allow categories suited to the approved statuses.
Evidence	Declaration, document, mandated check	Keep the verification level and its date.
Period	Active link, historical, disputed	The event is assessed at its own date, not at the entry date.
Aid duplicate	By event, person, household	Define the permitted national and local cumulations.
Access	Social officer, committee, treasurer	The treasurer sees amount and beneficiary, not the diagnosis.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behaviour
FAM01	Age is calculated at the date of the event; an uncertain date of birth blocks the thresholds concerned.
FAM02	The exact ten-year threshold, absent from the manual's below/above distinction, requires an explicit decision.
FAM03	A disputed family relationship must not be published in the general directory.
FAM04	A change of payee does not alter the identity of the person who establishes the right.

B · Form and data-entry experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced by fictitious values. Final validation runs server-side.

Functional field	Type or content	Validation and input assistance
D04-F01 · Related person	Minimum identity	Name, date of birth if necessary and internal identifier.
D04-F02 · Declaring member	Membership reference	Check belonging to the scope concerned.
D04-F03 · Link type	Configured list	Present the supporting documents required for the chosen type.
D04-F04 · Link start and end	Dates	Forbid an end earlier than the start.
D04-F05 · Verification state	Declared, checked, disputed	A change keeps author, date and evidence.
D04-F06 · Event	Type, date, reference	Search for requests already linked to the same event.
D04-F07 · Payment recipient	Person or third party	Distinguish beneficiary of the right from beneficiary of the payment.
D04-F08 · Confidentiality level	Classification	Apply masking, retention period and access log.

COMMON CONTEXT AND EDITING RIGHTS

The screen always shows the association, activity or file concerned, the currency where necessary, the status and the version. The author, creation date and identifier are generated by the system. The role authorised at the workflow step determines the available actions; a hidden button does not replace the access control.

> **Summary before validation.** The screen presents the inputs, the expected result and the errors before the final action. After submission, any significant correction displays the differences; errors target the field concerned and preserve the other entered data.

Screen actions: save a draft, verify, submit, request additional information, approve or reject according to role.

C · Workflow, exceptions and transactional control

Actor	Action	Control
Member	Declare the link	Minimum information
Social officer	Verify the evidence	Specific authorization
System	Search for duplicates	Person and event
Committee	Validate the right	Rule applicable on the date
Treasurer	Pay the recipient	Identity and reference
Data controller	Archive or purge	Retention policy

EXCEPTIONS AND RECOVERY

Trigger	Response	Expected trace or recovery
Unknown date of birth	Request evidence	Do not arbitrarily choose a bracket.
Two members declare the same child	Keep both links	Apply the per-event cumulation rule.
Disputed supporting document	Suspend the review	Organize a human review and a remedy.
Wrong recipient	Block the payment	Rectify before issue, without changing the right.

Trigger	Response	Expected trace or recovery
Erasure request	Assess each category	Keep the necessary evidence and minimize the other data.

D · Combinations, test recipes and indicators

COMBINATIONS AND QUALIFICATIONS

Combination of functions	Qualification	Validity condition
Local + national aid	Conditional	Two distinct rights, cumulation allowed and explicit reconciliation.
Household + individual cap	Conditional	Define the order of the caps before calculation.
Family + public directory	Restricted	Publication only on authorized choice.
Death + loan	Compatible	Handle succession and credit in separate files.
Third-party payment + confidentiality	Compatible	Transmit the minimum necessary for settlement.

TEST SCENARIOS

Case	Data or trigger	Verifiable result
D04-T01 · Ten-year-old child	Threshold not configured	The calculation requests an arbitration instead of producing zero.
D04-T02 · Two requests for one death	Same event and same fund	The duplicate is flagged before budget reservation.
D04-T03 · Treasurer access	Medical document present	The document is masked; the amount payable remains visible.
D04-T04 · Change of spouse	Old event	The relationship valid at that date is used.
D04-T05 · Member export	Solidarity report	No diagnosis or third-party document is included.

MANAGEMENT INDICATORS

Indicator	Formula	Reading
Event duplicates	Confirmed alerts / requests reviewed	Do not present an alert as proven fraud.
Sensitive accesses	Document reads by authorized actor	Detect bulk or out-of-mission consultations.

D05

DOMAIN 05 · OPERATIONAL SHEET

Governance, mandates and votes

Governance links bodies, seats, mandates, delegations and decisions. The software must compute a reproducible quorum, then a majority on an explicitly chosen basis. A majority of votes cast is not a majority of those entitled to vote. Abstentions, proxies, vacant seats and conflicts of interest change these bases according to the rules. Secret voting requires a separation between verification of the right to vote and storage of the ballot. Elections, ordinary motions, changes to the rules and dissolutions use different policies.

Manual reference: p. 14–16, 22, 49–51, 75. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Body	GA, board, committee, commission, council	Each body has its own members and competences.
Quorum	Number, proportion, first and second convening	Round the required minimum up.
Majority	Relative, absolute, qualified, unanimity	Fix denominator, ties and abstentions.
Ballot	Public, secret, single-name, list, ranking	Choose a verifiable mechanism suited to the ballot.
Mandate	Duration, renewal, substitution	Check concurrent holdings and the renewal limit.
Delegation	One-off proxy, functional authority	Limit duration, scope and sub-delegation.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behaviour
GOV01	Freeze the electoral roll and the ballot policy before opening, with a dispute procedure.
GOV02	The same voting right cannot be exercised both directly and by proxy.
GOV03	The interested person recuses themselves where the rules require it; the denominator must be recalculated explicitly.
GOV04	A technical administrator cannot manufacture a statutory decision or alter a closed result.

B · Form and data-entry experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced by fictitious values. Final validation runs server-side.

Functional field	Type or content	Validation and input assistance
D05-F01 · Body concerned	Body reference	Check competence over the subject of the decision.
D05-F02 · Resolution subject	Text and documents	The version put to the vote must remain frozen.
D05-F03 · Electoral base	Dated list	Include categories, exclusions and votes per member.
D05-F04 · Quorum	Numerator, denominator, rate	State whether proxies and remote participants are counted.
D05-F05 · Majority	Type and threshold	Define the treatment of abstentions, blank, spoilt and tied votes.
D05-F06 · Proxies	Grantor and proxy holder	Cap on mandates and no delegation cycle.
D05-F07 · Secret mode	Boolean and protocol	Separate proof of participation from the ballot.
D05-F08 · Result and minutes	Signed report	Publish the totals and the dispute channels.

COMMON CONTEXT AND EDITING RIGHTS

The screen always shows the association, activity or file concerned, the currency where necessary, the status and the version. The author, creation date and identifier are generated by the system. The role authorised at the workflow step determines the available actions; a hidden button does not replace the access control.

> **Summary before validation.** The screen presents the inputs, the expected result and the errors before the final action. After submission, any significant correction displays the differences; errors target the field concerned and preserve the other entered data.

Screen actions: save a draft, verify, submit, request additional information, approve or reject according to role.

C · Workflow, exceptions and transactional control

Actor	Action	Control
Secretary	Prepare the ballot	Competence and timetable
Committee	Freeze the electorate	Prior remedy
Chair of the meeting	Verify the quorum	Basis and attendees
Voters	Vote	Uniqueness of the right
Scrutineers	Count the votes	Totals and rules
Body	Proclaim and archive	Minutes and dispute deadline

EXCEPTIONS AND RECOVERY

Trigger	Response	Expected trace or recovery
Insufficient quorum	Suspend the decision	Prepare the second convening provided for.
Tie	Apply the declared rule	Invent neither a casting vote nor a draw.
Expired mandate	Withdraw the authority	Maintain the ordinary authorized consultations.
Interrupted vote	Place in incident	Resume or cancel according to the approved protocol.

Trigger	Response	Expected trace or recovery
Dispute	Freeze execution if required	Keep ballots, evidence and the version of the rules.

D · Combinations, test recipes and indicators

COMBINATIONS AND QUALIFICATIONS

Combination of functions	Qualification	Validity condition
Weighted vote + legal person	Conditional	Weights fixed and published before opening.
Secret vote + proxy	Conditional	The protocol must preserve secrecy and uniqueness.
Conflict of interest + loan	Compatible	Recusal of the applicant and recalculation of voters.
Section + national delegates	Compatible	Dated mandates and representation caps.
Offline + sensitive vote	Restricted	No improvised merging of contradictory ballots.

TEST SCENARIOS

Case	Data or trigger	Verifiable result
D05-T01 · Quorum 3/5 of 17	Ten present	Quorum not reached; eleven are required.
D05-T02 · Absolute majority of votes cast	10 for, 8 against, 3 abstentions excluded	The resolution passes if the policy requires more than 9 votes.
D05-T03 · Double vote	Grantor and proxy holder both try to vote	Only one participation is accepted.
D05-T04 · Mandate limited to two terms	Third candidacy	Refused with reasons unless an applicable statutory change applies.
D05-T05 · Secret ballot	Audit export	Proves participation without revealing how anyone voted.

MANAGEMENT INDICATORS

Indicator	Formula	Reading
Participation	Rights exercised / rights opened	Separate persons present from votes represented.
Decision lead time	Close of the vote minus complete file	Break down by body and type of resolution.

D06

DOMAIN 06 · OPERATIONAL SHEET

Meetings, calendar and attendance

The meeting brings the members together; the financial session carries an activity due date. Their association is optional: several activities can share one meeting and an activity can operate without a physical meeting. The calendar must handle a fixed day, the nth weekday, the first Saturday strictly after the 5th, custom periodicity and exceptional postponement. A change of date does not erase receivables already issued. The minutes record attendance, decisions and tasks, but financial details remain tied to their specialized journals.

Manual reference: p. 36–40, 49–51, 75. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Recurrence	Weekly, monthly, yearly, relative rule	Preview at least twelve occurrences before validation.
Format	In person, remote, hybrid	Define the accepted proof of participation.
Convening notice	Notice period specific to the body and the type	The 45-day FEV notice prevails over a generic 30-day council notice.
Attendance	Present, late, absent, excused, early departure	Keep times and the source of the observation.
Postponement	Same obligation, new date, cancellation	Explicitly choose the treatment of calls.
Minutes	Draft, reviewed, signed, published, corrected	A correction creates a linked version.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behaviour
REU01	A recurrence uses the association's local time zone and stores instants in UTC.
REU02	A calendar change must not generate a second contribution for the same due date.
REU03	Physical attendance does not amount to payment and a paid contribution does not amount to attendance.
REU04	Automatic fines remain contestable and attached to the initial observation.

B · Form and data-entry experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced by fictitious values. Final validation runs server-side.

Functional field	Type or content	Validation and input assistance
D06-F01 · Meeting type	Enumeration	Choose the body and the corresponding convening notice period.
D06-F02 · Calendar rule	Controlled expression	Show calculated dates and excluded days.
D06-F03 · Time zone	Zone identifier	Association value proposed; explicit exception.
D06-F04 · Venue or link	Protected text	Distribute the private link only to authorized participants.
D06-F05 · Agenda	Ordered list	Lock resolutions according to the convening rules.
D06-F06 · Attendance	Named grid	Bulk entry with confirmation of absences.
D06-F07 · Linked sessions	Activity references	A financial occurrence cannot be linked twice.
D06-F08 · Minutes actions	Owner and due date	Require an active owner, status and proof of completion.

COMMON CONTEXT AND EDITING RIGHTS

The screen always shows the association, activity or file concerned, the currency where necessary, the status and the version. The author, creation date and identifier are generated by the system. The role authorised at the workflow step determines the available actions; a hidden button does not replace the access control.

> **Summary before validation.** The screen presents the inputs, the expected result and the errors before the final action. After submission, any significant correction displays the differences; errors target the field concerned and preserve the other entered data.

Screen actions: save a draft, verify, submit, request additional information, approve or reject according to role.

C · Workflow, exceptions and transactional control

Actor	Action	Control
Secretary	Schedule	Recurrence previewed
Board	Convene	Notice period respected
Reception	Record attendance	Identities verified
Chair	Open the proceedings	Quorum observed
Secretary	Draft the minutes	Decisions and tasks
Signatories	Publish	Signed version archived

EXCEPTIONS AND RECOVERY

Trigger	Response	Expected trace or recovery
Public holiday	Propose a postponement	Require validation and a new notification.
Meeting cancelled	Keep the occurrence	Decide separately on maintaining the calls.
Attendance corrected	Create a correction	Recalculate the penalty by reversal entry.
Connection lost	Record as pending	No definitive remote attendance without an evidence rule.

Trigger	Response	Expected trace or recovery
Minutes disputed	Open a corrected version	Keep the original minutes and the approval.

D · Combinations, test recipes and indicators

COMBINATIONS AND QUALIFICATIONS

Combination of functions	Qualification	Validity condition
Meeting + loan session	Compatible	Due dates remain payable regardless of attendance.
Hybrid + quorum	Conditional	Define which remote participants are recognized.
Postponement + penalty	Conditional	Specify whether the financial due date is also postponed.
National + local calendar	Compatible	Detect conflicts without blocking authorized choices.
Refreshments + cancelled meeting	Conditional	Handle expenses already incurred and deposits.

TEST SCENARIOS

Case	Data or trigger	Verifiable result
D06-T01 · First Saturday after the 5th	The 5th falls on a Saturday	The calculated date is the 12th, if “after” is strict.
D06-T02 · Meeting with three funds	Tontine, solidarity, refreshments	Three separate calls and a single attendance.
D06-T03 · Postponement to the following month	Call already paid	No new debt for the same obligation.
D06-T04 · Excuse approved	Fine already generated	The waiver leaves a trace and reduces the debt.

Case	Data or trigger	Verifiable result
D06-T05 · Corrected minutes	Existing published version	The reader sees the new version and the link to the old one.

MANAGEMENT INDICATORS

Indicator	Formula	Reading
Attendance record	Recognized attendances / eligible convenings	Show excused absences separately.
Actions completed on time	Actions closed before due date / actions due	Do not confuse published minutes with completed action.

D07

DOMAIN 07 · OPERATIONAL SHEET

Discipline, sanctions and appeals

The disciplinary register separates finding, qualification, decision, receivable and appeal. A scale may set a fixed amount, a percentage or a progression, but it must explain how amounts combine. The SLD sanctions of 1 000, 3 000 and 5 000 FCFA for successive absences do not by themselves say whether each tier replaces or adds to the previous one. The target requires an explicit choice before activation. Non-financial sanctions, warnings or suspension, follow their own rights of defence and their own timetable. A debt waiver is never presented as a payment.

Manual reference: p. 37, 49–51. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Sanction	Warning, fine, suspension, exclusion	Decision authority and appeal specific to each level.
Base	Flat amount, rate, progressive amount	State whether the base is a debt, a contribution or another assessment base.
Combination	Additive, maximum, replacement, exclusive	Prevent double billing of the same breach.
Repeat offence	Consecutive, rolling period, financial year	An excuse may or may not interrupt the series depending on the rule.
Appeal	Suspensive or non-suspensive	The disputed debt appears separately.
Waiver	Full, partial, conditional	Retain authority, reason and impact on the fund.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behavior
SAN01	A single finding creates only one sanction per rule and version, even if the process is run again.
SAN02	A sanction must reference the article applicable on the date of the breach.
SAN03	The author of the finding does not alone rule on their own appeal where separation is required.
SAN04	Cancelled sanctions are reversed and remain in the audit.

B · Form and data-entry experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced by fictitious values. Final validation runs on the server.

Functional field	Type or content	Validation and data-entry assistance
D07-F01 · Member concerned	Membership reference	Check that the member belongs to the association on the date of the fact.
D07-F02 · Breach	Scale code	Display article, amount and combination before decision.
D07-F03 · Date and session	Date and reference	The session is optional for facts outside a meeting.
D07-F04 · Facts established	Text and documents	Distinguish observation, testimony and qualification.
D07-F05 · Repeat-offence series	Explainable calculation	Display the facts retained and the interruptions.
D07-F06 · Proposed amount	Monetary	Any departure from the scale requires a reason.
D07-F07 · Appeal	Text, date, documents	Display deadline and competent authority.
D07-F08 · Final decision	Status and signatory	Prohibit a waiver greater than the remaining debt.

COMMON CONTEXT AND EDITING RIGHTS

The screen always shows the association, activity or file concerned, the currency where necessary, the status and the version. The author, creation date and identifier are generated by the system. The role authorised at the workflow step determines the available actions; a hidden button does not replace the access control.

> **Summary before validation.** The screen presents the inputs, the expected result and the errors before the final action. After submission, any significant correction displays the differences; errors target the field concerned and preserve the other entered data.

Screen actions: save a draft, verify, submit, request additional information, approve or reject according to role.

C · Workflow, exceptions and transactional control

Actor	Action	Control
Censor	Record the fact	Evidence and date
Secretary	Notify	Article and appeal
Competent authority	Decide	Scale and recusal
Treasurer	Create the receivable	Approved amount
Appeals committee	Review	Reasoned decision
System	Apply the outcome	Consistent debt and journal

EXCEPTIONS AND RECOVERIES

Trigger	Response	Expected trace or recovery
Duplicate finding	Merge the references	Keep only one active financial obligation.
Late excuse	Review the series	Correct downstream fines if the rule requires it.
Ambiguous scale	Block automation	Ask the competent body for the combination mode.
Fine already paid then cancelled	Create a credit note	Refund or allocation with agreement.

Trigger	Response	Expected trace or recovery
Out-of-time appeal	Record admissibility	Do not erase the request or invent a derogation.

D · Combinations, test recipes and indicators

COMBINATIONS AND QUALIFICATIONS

Function combination	Qualification	Validity condition
Fine + solidarity	Conditional	Allocation to the fund defined, without mixing social rights and sanction.
Suspension + active loan	Compatible	Collection remains open.
Appeal + vote	Conditional	Voting rights set by the bylaws, not by an unpaid badge alone.
Progressive sanctions + excuses	Conditional	Recalculation of the series under the published rule.
Attendance correction + closing	Compatible	Adjustment dated within the open period.

TEST RECIPE SCENARIOS

Case	Data or trigger	Verifiable result
D07-T01 · SLD lateness	Scale 500 FCFA	Receivable 500 and no cash entry before payment.
D07-T02 · Two absences	Replacement mode configured	The total follows the chosen tier, with no implicit accumulation.
D07-T03 · Job re-run	Same finding and rule	No second sanction is created.
D07-T04 · Waiver of 300 on 500	Signed decision	Remaining due 200; collection equal to zero.
D07-T05 · Cancellation after payment	500 already collected	Credit note of 500 and refund trail.

STEERING INDICATORS

Indicator	Formula	Reading
Cancellation rate	Cancelled sanctions / sanctions decided	Analyze misunderstood rules, without humiliating ranking.
Disputed stock	Amount of sanctions under appeal	Separate what is immediately collectable from what is suspended.

D08

DOMAIN 08 · OPERATIONAL SHEET

Documents, internal rules and signatures

The documentary repository holds bylaws, internal rules, contracts, minutes, supporting documents, reports and templates. Every document has an identity, versions and a confidentiality level. Executable rules are linked to the articles that authorize them, so that a member understands why an amount is requested from them. Electronic signature is an extension: an uploaded file, an acceptance checkbox and a qualified signature must not be presented as equivalent. Archiving must preserve the original document and the validation evidence, regardless of the preview format.

Manual reference: p. 23, 51, 69–70, 82. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Type	Bylaws, internal rules, minutes, contract, receipt, document	Metadata and rights adapted to each category.
Version	Draft, in review, approved, published, replaced	A published version is immutable.
Approval	Author then reviewer, board, GA	The approval rule depends on the document.
Signature	Scanned handwritten, acceptance, external service	Display the level of proof actually available.
Distribution	Public, members, board, restricted folder	Download links must comply with the same rights.
Retention	Duration, litigation hold, controlled purge	Configure after validation of the applicable framework.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behavior
DOC01	The content fingerprint detects a modification; it does not by itself prove the signatory's identity.
DOC02	A new rule becomes active only on its effective date and after the required approval.
DOC03	Executable or infected documents are quarantined, outside public access.
DOC04	Deleting an account does not destroy the financial documents needed for the audit.

B · Form and data-entry experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced by fictitious values. Final validation runs on the server.

Functional field	Type or content	Validation and data-entry assistance
D08-F01 · Title and type	Text and list	Understandable title, without unnecessary sensitive data.
D08-F02 · Original file	Controlled binary	Format, size, file signature and antivirus scan.
D08-F03 · Version	Number and parent	The system assigns a non-reusable sequence.
D08-F04 · Effective date	Date	Distinguish adoption, publication and entry into force.
D08-F05 · Linked articles	Structured references	Link executable parameters to the relevant articles.
D08-F06 · Approvers	Roles and persons	Require mandates valid on the decision date.
D08-F07 · Classification	Access level	Control preview, export and sharing.
D08-F08 · Retention	Policy and hold	A purge suspension for litigation must be justified.

COMMON CONTEXT AND EDITING RIGHTS

The screen always shows the association, activity or file concerned, the currency where necessary, the status and the version. The author, creation date and identifier are generated by the system. The role authorised at the workflow step determines the available actions; a hidden button does not replace the access control.

> **Summary before validation.** The screen presents the inputs, the expected result and the errors before the final action. After submission, any significant correction displays the differences; errors target the field concerned and preserve the other entered data.

Screen actions: save a draft, verify, submit, request additional information, approve or reject according to role.

C · Workflow, exceptions and transactional control

Actor	Action	Control
Author	File	Format and classification
System	Check the file	Quarantine if risk
Reviewer	Verify the content	Articles and consistency
Authority	Approve	Vote or signatures
Secretary	Publish	Version and effective date
Archivist	Preserve	Fingerprint and proof

EXCEPTIONS AND RECOVERIES

Trigger	Response	Expected trace or recovery
Unreadable file	Reject publication	Keep the draft and request a valid file.
Confidential document misclassified	Remove sharing	Audit accesses already made.
Signature refused	Return to review	Do not mark the document as signed.
Contradictory text and parameter	Block activation	Have the interpretation arbitrated and record the decision.

Trigger	Response	Expected trace or recovery
Open litigation	Freeze the purge	Restrict access without destroying the evidence.

D · Combinations, test recipes and indicators

COMBINATIONS AND QUALIFICATIONS

Function combination	Qualification	Validity condition
Internal rules + rule engine	Compatible	Article-parameter link and prior simulation.
Signature + offline	Conditional	Collect a draft, confirm the proof online.
Medical document + global export	Prohibited	Export an aggregate without nominative documents.
Template + customization	Compatible	Preserve provenance and local differences.
Archiving + dissolution	Compatible	Maintain an owner and consultation access.

TEST RECIPE SCENARIOS

Case	Data or trigger	Verifiable result
D08-T01 · Rules replaced	Two approved versions	Old operations point to the old rule.
D08-T02 · Expired link	Download of a private document	Access denied even if the file path is known.
D08-T03 · Same-name file	Same name, different content	New version and different fingerprint.
D08-T04 · Expired mandate	Approval attempted	Refusal before publication.
D08-T05 · Scheduled purge	Document under litigation hold	The purge is suspended and logged.

STEERING INDICATORS

Indicator	Formula	Reading
Complete documents	Files with mandatory documents / active files	Measure by type of procedure.
Traceable rules	Active parameters linked to a decision / parameters subject to validation	Identify parameters without documented authority.

D09

DOMAIN 09 · OPERATIONAL SHEET

Rule engine and activity contracts

An activity is a parameterized, versioned and dated contract. The engine combines participation, calculation, eligibility, financing and validation rules. A complete configuration cannot be reduced to a free JSON object: its schema, units, dependencies and incompatibilities must be checked before publication. The engine produces a detailed explanation of each decision and allows simulation on anonymized data. A change may apply to new operations, to a new period or to an accepted amendment; it must not rewrite historical amounts.

Manual reference: p. 27–35, 65–66. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Scope	Platform, country, association, activity, contract	The most specific rule applies only if the override is authorized.
Timing	Publication, effect, expiry	Separate decision date and economic date.
Expression	Flat amount, percentage, brackets, minimum, maximum	Typed expressions without arbitrary code execution.
Eligibility	AND, OR, exclusions, derogations	Display the precise reason for refusal and the appeal channel.
Derogation	Prohibited, justified, double approval	The derogation applies to one case file, not to the whole activity.
Simulation	Nominal case, limits, replayed history	Compare the old and the new rule before activation.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behavior
REG01	Every operation keeps the rule version, the calculation inputs and the rounded result.
REG02	A blocking incompatibility prevents publication, even if each field is valid in isolation.
REG03	Monetary calculations use exact decimals or integer units, never binary floating point.
REG04	A configurable engine keeps non-deactivatable invariants: isolation, uniqueness, audit and balance.

B · Form and data-entry experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced by fictitious values. Final validation runs on the server.

Functional field	Type or content	Validation and data-entry assistance
D09-F01 · Activity code	Stable identifier	Unique within the association, distinct from the editable label.
D09-F02 · Product type	Versioned catalogue	Determines the schemas and rules available.
D09-F03 · Participation	Mandatory or voluntary	Define categories, entry date and exit date.
D09-F04 · Calculation expression	Typed expression tree	Validate currency, duration, rate and authorized references.
D09-F05 · Financing mode	Funds and reserves	Distinguish liquidity from the theoretical cap.
D09-F06 · Validation policy	Thresholds and roles	Prohibit self-approval at sensitive steps.
D09-F07 · Effective date	Date and version	Prevent effects on contracts already active.
D09-F08 · Simulation set	Cases and expected results	Publication requires passing the mandatory tests.

COMMON CONTEXT AND EDITING RIGHTS

The screen always shows the association, activity or file concerned, the currency where necessary, the status and the version. The author, creation date and identifier are generated by the system. The role authorised at the workflow step determines the available actions; a hidden button does not replace the access control.

> **Summary before validation.** The screen presents the inputs, the expected result and the errors before the final action. After submission, any significant correction displays the differences; errors target the field concerned and preserve the other entered data.

Screen actions: save a draft, verify, submit, request additional information, approve or reject according to role.

C · Workflow, exceptions and transactional control

Actor	Action	Control
Manager	Compose the rules	Valid schema
System	Detect conflicts	Dependencies and units
Controller	Simulate	Limit cases covered
Board	Approve	Decision and scope
System	Publish on the effective date	Immutable version
Auditor	Replay a calculation	Reproducible result

EXCEPTIONS AND RECOVERIES

Trigger	Response	Expected trace or recovery
Division by zero	Refuse the calculation	Display the missing input and create no movement.
Two rules of the same priority	Block publication	Require a priority or an explicit exclusion.
Retroactive amendment	Open a correction	Never rewrite closed entries.
Unknown parameter on import	Quarantine	Keep the value and request a mapping.

Trigger	Response	Expected trace or recovery
Expired derogation	Return to the normal rule	The contract keeps operations already authorized.

D · Combinations, test recipes and indicators

COMBINATIONS AND QUALIFICATIONS

Function combination	Qualification	Validity condition
Local rule + national baseline	Conditional	Override only on free parameters.
Fractions + integer distribution	Conditional	Grouping contract or explicit fractional distribution.
Loan + capitalization	Conditional	Consent, formula and validation of the applicable framework.
Derogation + reserved fund	Conditional	The derogation does not create liquidity.
Shared template + update	Compatible	Voluntary migration with version comparison.

TEST RECIPE SCENARIOS

Case	Data or trigger	Verifiable result
D09-T01 · Rate without periodicity	Value 3 % alone	Publication refused until month or year is defined.
D09-T02 · Change of share	Cycle already open	Amendment required and impact presented per member.
D09-T03 · Simulation of 100 case files	New cap rule	Differences listed before any real entry.
D09-T04 · Old contract	Rule replaced	The recalculation uses its historical version.
D09-T05 · Dangerous configuration	Audit disabled	The platform refuses this non-derogable parameter.

STEERING INDICATORS

Indicator	Formula	Reading
Explainable calculations	Calculations retaining inputs and version / calculations executed	Design target: 100 % of financial calculations.
Rejected configurations	Conflicts by type of parameter	Improve templates from frequent errors.

D10

DOMAIN 10 · OPERATIONAL SHEET

Contributions, calls and allocations

An expected contribution is a receivable; a collection is a cash movement; an allocation links the two. This separation allows partial payments, advances, grouped payments, category changes and corrections. Calls may be individual, per household, per share, progressive or proportional to a declared base. Statuses must be orthogonal: a debt can be both partially paid and overdue. The software therefore keeps settlement status, due-date status and dispute status, instead of a single ambiguous status.

Manual reference: p. 31, 36–40, 75–76. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Base	Member, household, share, declared income, bracket	Keep the base and its period of validity.
Frequency	One-off, session, month, year, event	A due-date key prevents duplicate calls.
Pro rata	None, actual days, month started	Define entry and exit during the period.
Payment	Full, instalment, advance, grouped	An unallocated amount remains an identifiable credit.
Exemption	Full, partial, temporary	Create an approved waiver rather than a false receipt.
Priority	Age, activity, member’s choice	Display the allocation order before confirmation.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behavior
COT01	Net due = initial call + debit adjustments - waivers - valid allocations.
COT02	Any overpayment becomes a credit; the remaining amount due displayed does not become negative.
COT03	Uniqueness applies to member, activity, due date and generation version.
COT04	A payment for several persons keeps the actual payer and the beneficiaries of each allocation.

B · Form and data-entry experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced by fictitious values. Final validation runs on the server.

Functional field	Type or content	Validation and data-entry assistance
D10-F01 · Called population	Member filter	Freeze the members retained and the exclusions.
D10-F02 · Period and due date	Dates	Specify grace period, time zone and postponement of non-working days.
D10-F03 · Calculation base	Shares or base	Read the subscription valid on the contractual date.
D10-F04 · Applicable rate	Version reference	Display unit amount and detailed formula.
D10-F05 · Exemption	Reason and decision	Reduce the receivable without inflating collections.
D10-F06 · Amount received	Monetary	Do not exceed the receipt available to allocate.
D10-F07 · Allocations	Line table	Allocated sum less than or equal to the validated payment.
D10-F08 · Remainder	Debt or credit	Present residual debt and available advance separately.

COMMON CONTEXT AND EDITING RIGHTS

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> **Summary before validation.** The screen presents the inputs, the expected result and the errors before the final action. After submission, any significant correction displays the differences; errors target the field concerned and preserve the other entered data.

Screen actions: save a draft, verify, submit, request additional information, approve or reject according to role.

C · Workflow, exceptions and transactional control

Actor	Action	Control
Manager	Prepare the calls	Population and rate
Controller	Validate the batch	Totals and exclusions
System	Issue	Unique keys
Treasurer	Collect	Proof of payment
Member or agent	Allocate	Confirmed priority
System	Update rights	Debt and lateness kept separate

EXCEPTIONS AND RECOVERIES

Trigger	Response	Expected trace or recovery
Member joined mid-month	Apply the pro rata	Display the days or months retained.
Payment without reference	Place on hold	Look for the payer before allocation.
Session cancelled	Cancel or postpone the call	Do not delete the payment already received.
Payment greater than the amount due	Create an advance	Offer a refund or the next due date.

Trigger	Response	Expected trace or recovery
Late exemption	Create a credit note	Keep the decision and the initial receipt.

D · Combinations, test recipes and indicators

COMBINATIONS AND QUALIFICATIONS

Function combination	Qualification	Validity condition
Contribution + tontine	Compatible	Eligibility reads the allocations of the right cycle.
Advance + new period	Compatible	Explicit allocation according to the member’s authorization.
In-kind + monetary contribution	Conditional	Approved valuation without inventing liquidity.
Family call + individual rights	Conditional	Define the members covered by the payment.
Pro rata + progressive rate	Conditional	Choose the order of the pro rata and the brackets.

TEST RECIPE SCENARIOS

Case	Data or trigger	Verifiable result
D10-T01 · Call 25 000, receipt 10 000	Due date passed	Due 15 000; partial and late statuses simultaneously.
D10-T02 · Payment 30 000	Debt of 25 000	Allocation 25 000, unallocated credit 5 000.
D10-T03 · Batch run twice	Same due-date key	A single receivable per member.
D10-T04 · Two members, one payer	Payment of 50 000	Two allocations of 25 000 linked to the same receipt.
D10-T05 · Full exemption	Call of 15 000	Due nil, collected nil, waiver 15 000.

STEERING INDICATORS

Indicator	Formula	Reading
Cohort collection	Allocations on period calls / net period calls	Do not mix old arrears with new calls.
Unallocated credits	Sum of validated payments not applied	Track age and presumed beneficiary.

D11

DOMAIN 11 · OPERATIONAL SHEET

Payments received and reconciliation

The manual distinguishes payment entry from operator integrations still to be finalized. The target treats each payment as a stateful operation, with an idempotency key, an external reference, a gross amount, fees and a net amount. A success notification in the interface is not enough to prove settlement. The confirmed state comes from verifiable evidence, then reconciliation compares the internal journal with the operator or bank statement. Refunds, cancellations and chargebacks create linked operations. An internal wallet is an accounting position; its qualification and its ability to hold funds must be validated separately.

Manual reference: p. 39–40, 52–59, 72, 82. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Channel	Cash, bank, Mobile Money, card via provider	Each connector declares its capabilities and its countries.
State	Initiated, pending, confirmed, failed, expired, reversed	Authorized transitions are controlled on the server.
Fees	Payer, association, shared	Keep gross, fees and net with the rule applied.
Reconciliation	Exact automatic, assisted, manual	Ambiguous matches require validation.
Refund	Full, partial, original channel	The cumulative refunded amount does not exceed the amount available.
Evidence	Numbered receipt, statement, signed callback	A screenshot alone does not confirm an integrated payment.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behavior
PAY01	The same provider reference cannot be posted twice within its scope.
PAY02	A wait timeout or a network error does not mean a definitive financial failure.
PAY03	Fees do not silently reduce the contribution settled; the rule must be explicit.
PAY04	A refund keeps the initial payment, its cancelled allocations and the reversing entries.

B · Form and data-entry experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced by fictitious values. Final validation runs on the server.

Functional field	Type or content	Validation and data-entry assistance
D11-F01 · Actual payer	Person or third party	May differ from the member whose debt is settled.
D11-F02 · Channel and account	Cash desk or provider reference	Active account and compatible currency.
D11-F03 · Gross amount	Positive monetary amount	Specify the unit and the channel limits.
D11-F04 · Fees	Amount and bearer	Check gross = net received + fees withheld where fees are deducted.
D11-F05 · External reference	Text unique per provider	Normalize without losing the original value.
D11-F06 · Idempotency key	Client identifier	A key reused with another amount is rejected.
D11-F07 · Value date	Operator date	Distinct from the entry date and the confirmation date.
D11-F08 · Evidence and reconciliation	Document or signed message	Keep the verified signature, the statement batch and the controller.

COMMON CONTEXT AND EDITING RIGHTS

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C · Workflow, exceptions and transactional control

Actor	Action	Control
Payer	Initiate	Amount and allocation
Provider or cashier	Receive	Authenticated proof
System	Confirm once	Idempotency and currency
Treasurer	Allocate	Debt or advance
Controller	Reconcile	External statement
System	Issue the receipt	Number and detail

EXCEPTIONS AND RECOVERIES

Trigger	Response	Expected trace or recovery
Duplicate webhook	Acknowledge receipt	Return the existing result without a new entry.
Out-of-order webhook	Check the current state	Do not move from confirmed back to pending.
External success after expiry	Reconcile	Confirm or refund according to policy, without losing the funds.
Amount discrepancy	Block automatic allocation	Create a discrepancy case file with evidence.

Trigger	Response	Expected trace or recovery
Chargeback	Reverse the settlement	Reopen the debts concerned according to the rule and notify.

D · Combinations, test recipes and indicators

COMBINATIONS AND QUALIFICATIONS

Function combination	Qualification	Validity condition
Mobile Money + offline	Restricted	Prepare an intention, confirm only on server-side proof.
Grouped payment + fees	Conditional	Split according to a constant rule and a preserved sum.
Multi-currency + contribution	Conditional	Rate, date and fees visible before validation.
Cash + digital receipt	Compatible	The receipt carries the cashier and the cash check.
Cancellation + closed session	Compatible	Adjustment in the open period with a historical link.

TEST RECIPE SCENARIOS

Case	Data or trigger	Verifiable result
D11-T01 · Two identical callbacks	Payment of 25 000	One receipt and a single increase in the cash desk.
D11-T02 · Gross 10 000, fees 100	Fees borne by the association	Debt settled 10 000, net cash 9 900, expense 100.
D11-T03 · Timeout then success	Same payment key	The member must not pay a second time.
D11-T04 · Partial refund	3 000 out of 10 000	Cumulative amount still available 7 000.
D11-T05 · Forged callback	Incorrect signature	No confirmation and no allocation.

STEERING INDICATORS

Indicator	Formula	Reading
Reconciliation rate	Reconciled payments / confirmed payments	Present count and amount, by channel.
Confirmation delay	Confirmation minus initiation	Publish median and 95th percentile, not only the mean.

D12

DOMAIN 12 · OPERATIONAL SHEET

Cash accounts, budgets and entries

The fund represents an economic allocation, the account represents the place where funds are held, and the budget represents an authorization to spend. A single bank may hold several funds; several accounts may serve the same fund. The target introduces a balanced general ledger, analytical dimensions and commitment reservations. It extends the manual's management reports without presuming full accounting compliance. Availability takes into account confirmed payments, reserved commitments, unavailable funds and pending operations.

Manual reference: p. 52–55, 57, 79. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Account	Cash, bank, provider, transit	A reconciliation specific to each account.
Fund	Free, allocated, reserved, refundable deposit	Transferring funds requires explicit authority.
Budget	Annual, project, activity, revised	Compare authorized, committed, paid and available.
Accounting	Management accounting, then target balanced journal	Mapping to the chosen framework after specialized validation.
Closing	Monthly, annual, per cycle	Lock the period and correct in an open period.
Validation	Thresholds, dual control, delegation	Check availability and separation of duties.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behavior
FIN01	For each accounting voucher and bookkeeping currency, total debit = total credit.
FIN02	A transfer movement between accounts is neither income nor expense in itself.
FIN03	A reservation reduces the available amount without changing the cash balance.
FIN04	No expense may consume a fund allocated to another purpose without an authorized decision.

B · Form and data-entry experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced by fictitious values. Final validation runs on the server.

Functional field	Type or content	Validation and data-entry assistance
D12-F01 · Voucher and type	Identifier and category	Stable numbering, not reused after cancellation.
D12-F02 · Economic date	Date	Open period; entry date kept separately.
D12-F03 · Cash account	Reference	Currency and reconciliation mandatory.
D12-F04 · Analytical fund	Activity or reserve reference	Prohibit the absence of a fund on an allocated movement.
D12-F05 · Debit and credit lines	Monetary table	Equal sums; active accounts; authorized precision.
D12-F06 · Associated budget	Line and version	Check the committed remainder and the overspend threshold.
D12-F07 · Supporting document	Document	Mandatory depending on type and amount.
D12-F08 · Approval	Roles and signatories	Two distinct persons where policy requires it.

COMMON CONTEXT AND EDITING RIGHTS

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Screen actions: save a draft, verify, submit, request additional information, approve or reject according to role.

C · Workflow, exceptions and transactional control

Actor	Action	Control
Requester	Commit an expense	Purpose and budget
Budget manager	Reserve	Sufficient available amount
Validator	Approve	Authority and document
Treasurer	Settle	Account and proof
Accountant	Post	Balanced voucher
Controller	Reconcile and close	Discrepancies resolved

EXCEPTIONS AND RECOVERIES

Trigger	Response	Expected trace or recovery
Insufficient budget	Block or request a revision	Do not automatically increase the authorization.
Positive bank balance, empty fund	Refuse the expense from the fund	Overall liquidity does not replace the allocation.
Cash discrepancy	Create an incident	Contradictory inventory before regularization.
Closed period	Post an adjustment	Reference the original voucher.

Trigger	Response	Expected trace or recovery
Failed disbursement	Release or maintain the reservation	Decision according to the planned recovery and expiry.

D · Combinations, test recipes and indicators

COMBINATIONS AND QUALIFICATIONS

Function combination	Qualification	Validity condition
Single fund + several activities	Conditional	Analytical dimensions mandatory.
Budget + loan commitment	Compatible	Reserve from the decision, before payment.
Cash + in-kind stock	Distinct	Stock cannot finance a disbursement without a sale.
Consolidation + internal transfer	Compatible	Eliminate the identified reciprocal flows.
Multi-currency + journal	Conditional	Keep original and converted amounts.

TEST RECIPE SCENARIOS

Case	Data or trigger	Verifiable result
D12-T01 · Transfer of 100 000 between cash desks	Two accounts of the same fund	Total cash unchanged and no income created.
D12-T02 · Reservation 80 000	Free balance 100 000	Available 20 000, accounting balance 100 000.
D12-T03 · Unbalanced voucher	Debit 10 000, credit 9 999	Rejected before validation.
D12-T04 · Reserved bereavement fund	Loan requested against this fund	Refused without a suitable statutory authorization.
D12-T05 · Correction after closing	November error	Adjustment dated in the open period, history preserved.

STEERING INDICATORS

Indicator	Formula	Reading
Prudent available balance	Confirmed free cash - reserved commitments	Exclude receivables and illiquid assets.
Reconciliation discrepancy	Journal balance - adjusted external balance	Detail operations in transit and real anomalies.

D13

DOMAIN 13 · OPERATIONAL SHEET

Tontine share subscriptions and fractions

The tontine contract must reconcile periodic contribution, beneficiary entitlement, duration and number of distributions. Financial shares are not voting seats. Fractions may be represented by whole quarter units or by controlled rational numbers, so as to avoid rounding errors. A subscription specifies how much the member pays in, how much they may receive and which obligations still bind them after receiving. Entering or leaving mid-cycle constitutes a financial amendment, not a simple profile change.

Manual reference: p. 29–35. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Unit	Whole share, half share, quarter, custom grid	The fraction grid is validated when the cycle opens.
Subscription	Fixed, multiple shares, capped	Check total commitments and concentration.
Contribution	Share × periodic rate, scale	Frequency is part of the contract.
Entitlement	Fixed beneficiary share or proportional pot	Do not mix the two models without a hybrid rule.
Transfer	Prohibited, approved assignment, inheritance	Transfer entitlements and obligations with consent.
Amendment	Next session, new cycle, approved recalculation	Present the impact on all participants.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behaviour
PAR01	Sum of distributed entitlements + remaining entitlements = contractual entitlements, after traced amendments.
PAR02	An early beneficiary remains liable for the scheduled future contributions.
PAR03	A fixed beneficiary amount requires a funding plan compatible with the number of sessions.
PAR04	A fraction is not rounded up to a whole share in order to gain a distribution advantage.

B · Form and data-entry experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced by fictitious values. Final validation runs server-side.

Functional field	Type or content	Validation and input assistance
D13-F01 · Cycle	Open cycle reference	Subscription prohibited on a closed cycle.
D13-F02 · Subscribing member	Active membership	Check eligibility and concentration limit.
D13-F03 · Number of shares	Grid decimal	Example: exact multiples of 0,25; no free-form value.
D13-F04 · Contribution rate	Amount per share and period	Display the forecast total commitment.
D13-F05 · Beneficiary entitlement	Quantity and reference amount	Distinguish gross entitlement from any fees.
D13-F06 · Order or preference	Rank or choice	A preference does not constitute a guaranteed award.
D13-F07 · Guarantees after benefit	References	Size against future obligations where the rules so provide.
D13-F08 · Contract acceptance	Date and proof	Keep the version, the schedule and the exit arrangements.

COMMON CONTEXT AND EDITING RIGHTS

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C · Workflow, exceptions and transactional control

Actor	Action	Control
Member	Choose the shares	Commitment simulation
Manager	Check the limits	Status and concentration
System	Check the balance	Contributions and entitlements
Member	Accept the contract	Version retained
Board	Open the cycle	Frozen population
System	Generate the calls	Exact quantities

EXCEPTIONS AND RECOVERY

Trigger	Response	Expected trace or recovery
Unauthorised fraction	Reject the entry	Offer the grid values.
New share after opening	Require an amendment	Do not move beneficiaries silently.
Assignment after benefit	Maintain the initial debt unless an approved takeover applies	Avoid transferring only the favourable entitlements.
Exit before benefit	Calculate the settlement	Deductions and refund per the accepted contract.

Trigger	Response	Expected trace or recovery
Death of the subscriber	Freeze automatic allocation	Handle debt, guarantee and rightful claimant separately.

D · Combinations, test recipes and indicators

COMBINATIONS AND QUALIFICATIONS

Function combination	Qualification	Validity condition
Fractions + multiple beneficiaries	Compatible	Distribute the minimum units and respect remaining entitlements.
Multiple shares + single turn	Conditional	Define grouped receipt or several ranks.
Short cycle + fixed amount	Conditional	The budget must fund all entitlements.
Assignment + surety	Conditional	The surety must accept substitution where necessary.
Savings + tontine shares	Distinct	Two entitlements and two ledgers, same member.

TEST SCENARIOS

Case	Data or trigger	Verifiable result
D13-T01 · Half share at 100 000	Periodic rate	Call of exactly 50 000.
D13-T02 · Four members with two shares and four with one	Total subscriptions	Twelve shares, not eight.
D13-T03 · Distribution of two shares out of twelve	Cycle by exhaustion	Entitlement progression 16,67 %, independently of the sessions.
D13-T04 · Transfer of one share	Assignor already a beneficiary	No new entitlement is created.
D13-T05 · Rate increase	Prior contract still active	Amendment required before any new debt.

STEERING INDICATORS

Indicator	Formula	Reading
Contractual coverage	Planned resources / promised distributions and planned charges	A ratio below one blocks opening without explicit funding.
Concentration	Shares of the largest member / total shares	Measure dependence on a single contributor.

D14

DOMAIN 14 · OPERATIONAL SHEET

Allocations, rotations and draws

The allocation engine selects available entitlements, then reserves the funds, then triggers a payment. An allocation is therefore not a disbursement. The manual states that full automation of eligibility and bidding remains to be completed: this sheet describes the target. Rotation, one-off draw, calendar draw, justified need and merit priority each require different evidence. The list of eligible members is frozen for the calculation, but a final check prevents payment to a member who has become ineligible under the applicable rules.

Manual reference: p. 31–35, 78, 82. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Rotation	Initial order, rotation by share, exchangeable rank	Rank changes are accepted and recorded in the history.
Draw	Session, full calendar, weighting by entitlement	Publish the algorithm and keep the proof of the draw.
Need	Committee, criteria, urgency priority	Grounds and recusal of interested persons.
Distribution	One or more shares per session	The session cap and remaining entitlements limit the result.
Shortfall	Blocking, splitting, deferral, authorised advance	No option creates fictitious funds.
Eligibility	Contributions, status, guarantees, previous receipts	Display exclusions and the rule version applied.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behaviour
ATT01	Allocated quantity limited by the fundable pot, the session cap and remaining entitlements.
ATT02	Reservation and consumption of entitlements are atomic against concurrent validations.
ATT03	A rejected draw cannot be repeated to favour a participant; cancellation requires a stated reason.
ATT04	A failed payment leaves an allocation pending settlement or explicitly cancelled, never a fictitious success.

B · Form and data-entry experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced by fictitious values. Final validation runs server-side.

Functional field	Type or content	Validation and input assistance
D14-F01 · Session and cycle	References	Check opening, currency and active policy.
D14-F02 · Available pot	Server calculation	Confirmed receipts plus deferrals minus reservations and authorised charges.
D14-F03 · Eligible population	Dated snapshot	Display inclusions, exclusions and remaining entitlements.
D14-F04 · Allocation mode	Enumeration	Mandatory parameters depend on the mode.
D14-F05 · Proposed quantity	Share units	Respect the minimum fraction and the caps.
D14-F06 · Proof of selection	Rank, draw or decision	Keep data sufficient for an independent review.
D14-F07 · Validators	Authorised persons	Prohibit self-validation where the beneficiary sits on the board.
D14-F08 · Payment destination	Verified account	Check the match with the authorised beneficiary.

COMMON CONTEXT AND EDITING RIGHTS

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C · Workflow, exceptions and transactional control

Actor	Action	Control
System	Calculate the pot	Confirmed funds
Manager	Freeze the eligible list	Exclusion reasons
Engine or committee	Select	Proof of the mode
Validator	Reserve the entitlements	Concurrent control
Treasurer	Disburse	Payment proof
System	Publish the result	Updated entitlements and balances

EXCEPTIONS AND RECOVERY

Trigger	Response	Expected trace or recovery
Person has become ineligible	Suspend the payment	Re-examine under the rule without quietly running the draw again.
Two simultaneous validations	Lock the resources	Only one consumes the same entitlement.
Insufficient pot	Apply the contractual option	Notify the shortfall and the next step.
Payment rejected	Keep the debt owed to the beneficiary	Retry with the same operation identity.

Trigger	Response	Expected trace or recovery
Draw error	Cancel with proof	Keep the result, the reason and the decision to start again.

D · Combinations, test recipes and indicators

COMBINATIONS AND QUALIFICATIONS

Function combination	Qualification	Validity condition
Draw + merit	Conditional	Define pre-selection then draw, with no hidden criterion.
Rotation + urgency	Conditional	Exchange or derogation approved by the parties concerned.
Fraction + fixed pot	Conditional	Half-entitlement amount calculated explicitly.
Multiple beneficiaries + cap	Compatible	Sum of the quantities checked at session level.
Advance + distribution	Conditional	Lender, debt, cost and repayment identified.

TEST SCENARIOS

Case	Data or trigger	Verifiable result
D14-T01 · Pot 650 000, entitlement 300 000	Two-share cap	Two shares at most and 50 000 deferred.
D14-T02 · Pot 250 000	Fixed entitlement 300 000, no fraction	No payable allocation.
D14-T03 · Member with 0,5 share remaining	Request for one share	Refusal of the excess quantity.
D14-T04 · Two treasurers validate	Same entitlement	One operation succeeds, the other receives an explicit conflict.
D14-T05 · Failed transfer	Approved allocation	The entitlement is not paid; the amount stays reserved according to policy.

STEERING INDICATORS

Indicator	Formula	Reading
Entitlements served	Shares paid / contractual shares	Also present shares allocated but not paid.
Distribution delays	Payments after the promised date / expected payments	Break down cash shortfall and channel incident.

D15

DOMAIN 15 · OPERATIONAL SHEET

Auctions, discounts and sharing of the proceeds

Auctions require a more precise contract than a percentage of a starting price. The SLD manual mentions a starting price of 5 % of the contribution rate; that expression does not allow the conclusion that the rate applies to the pot. The target imposes a named base: contribution, beneficiary entitlement, pot or fixed amount. The winning offer may be a monetary discount, a discount rate or a separate cost. Offers are time-stamped by the server, ties and extensions are deterministic, and the amount retained is allocated to an identified fund.

Manual reference: p. 32–35, 75–76, 82. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Offer	Discount amount, rate, additional cost	An auction uses a single published unit.
Base	Pot, gross entitlement, contribution, flat fee	Prohibit percentages without a base.
Procedure	Open ascending, sealed bids, single-round auction	Choose the visibility and closing rule.
Tie	First valid offer, draw, new round	Publish the rule before offers are received.
Retention	Reserve fund, proceeds to be shared, fees	Distinguish retention of principal from distributable proceeds.
Participation	Eligible members, guarantee, offer cap	Prevent an offer above the authorised net entitlement.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behaviour
ENC01	Net amount = gross entitlement - discount - fees borne by the beneficiary, under an explicit contract.
ENC02	The discount remains in the treasury; its allocation must not be counted as a second receipt.
ENC03	The winning offer is irrevocable once validated under the accepted conditions.
ENC04	The administrator cannot change the receipt time or erase a losing offer.

B · Form and data-entry experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced by fictitious values. Final validation runs server-side.

Functional field	Type or content	Validation and input assistance
D15-F01 · Lot put up for auction	Entitlement or number of shares	Display gross entitlement, currency and number of winners.
D15-F02 · Mode and visibility	Enumeration	Define who may see offers before closing.
D15-F03 · Discount base	Code and amount	Value frozen at opening; any change requires cancellation.
D15-F04 · Starting price	Amount or rate	Unit and worked example displayed to the participant.
D15-F05 · Minimum increment	Positive amount	Validate offers against an exact grid.
D15-F06 · Closing time	Server instant	State any extension and the time zone displayed.
D15-F07 · Member’s offer	Amount or rate	Check entitlements, cap and capacity to commit.
D15-F08 · Allocation of the retention	Fund and key	Publication before the auction and traceable accounting.

COMMON CONTEXT AND EDITING RIGHTS

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C · Workflow, exceptions and transactional control

Actor	Action	Control
Manager	Publish the lot	Base and rules
Members	Submit offers	Eligibility and time-stamping
System	Close	Time and extension
Scrutineer	Verify the result	Ties and exclusions
Treasurer	Pay the net amount	Discount allocated
Controller	Review the auction	Complete audit trail

EXCEPTIONS AND RECOVERY

Trigger	Response	Expected trace or recovery
No offer	Apply the planned fallback	Deferral or rotation, with no arbitrary allocation.
Equal offers	Execute the published rule	Keep the proof of the tie-break.
Offer after closing	Reject	Server time is authoritative in the protocol.
Winning bidder defaults	Open an incident	Guarantee, new round or next bidder per contract.

Trigger	Response	Expected trace or recovery
Incorrect base	Cancel before payment	Inform all participants and reopen on a new version.

D · Combinations, test recipes and indicators

COMBINATIONS AND QUALIFICATIONS

Function combination	Qualification	Validity condition
Auction + fractions	Conditional	Define a split lot and the rate base for each lot.
Auction + loans	Conditional	Do not disguise a discount as a loan rate without qualification.
Auction + solidarity	Compatible	Retention allocation authorised and published.
Auction + offline	Restricted	No definitive offer without a time stamp accepted by the server.
Auction + multiple beneficiaries	Conditional	Define a uniform price or a price per winner.

TEST SCENARIOS

Case	Data or trigger	Verifiable result
D15-T01 · Entitlement 600 000, discount 30 000	No fees	Payment 570 000 and retention 30 000, total 600 000.
D15-T02 · 5 % on 25 000	Contribution base	Minimum 1 250, not 30 000.
D15-T03 · 5 % on 600 000	Gross entitlement base	Minimum 30 000; configuration different from the previous case.
D15-T04 · Two offers at the same amount	First-offer rule	Winner determined by a valid server time stamp.

Case	Data or trigger	Verifiable result
D15-T05 · Repeated payment webhook	Auction already settled	No second disbursement and no second proceeds entry.

STEERING INDICATORS

Indicator	Formula	Reading
Weighted average discount	Sum of discounts / sum of gross entitlements	Display volume, period and auction mode.
Effective competition	Distinct valid offers per lot	A lot without competition must be visible to control.

D16

DOMAIN 16 · OPERATIONAL SHEET

Tontine cycles, closings and defaults

A cycle may end after a number of sessions, on a fixed date, or once all entitlements have been served. These conditions do not guarantee that debts are settled: operational closing and financial closing must be kept distinct. The engine calculates remaining entitlements, future obligations, cash advances and the result to be allocated. A default before benefit and a default after benefit do not carry the same risk. The SLD uplifts of 15 % and 20 % remain local parameters whose base, periodicity and cap must be decided.

Manual reference: p. 31–35, 78–81. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
End	Fixed number, fixed date, exhaustion of entitlements	Define what happens if the financial objective is not reached.
Surplus	Deferral, reserve, refund, sharing	The allocation rule is accepted before opening.
Deficit	Supplementary call, advance, approved reduction	Do not close by wiping out remaining entitlements.
Default	Before benefit, after benefit, partial	Guarantees and procedures suited to the risk.
Closing	Operational, financial, archiving	Residual receivables remain tracked.
Renewal	Same model, new model, no carry-over	Balances are carried over through explicit entries.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behaviour
CYC01	The closing reconciles receipts, distributions, deferrals, fees and transfers; no opaque adjustment.
CYC02	An unserved entitlement appears in the final statement and receives an approved treatment.
CYC03	The obligations of a defaulting beneficiary are not cancelled by the closing of the sessions.
CYC04	The balance of the new cycle comes from an identified transfer, not from a manually edited field.

B · Form and data-entry experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced by fictitious values. Final validation runs server-side.

Functional field	Type or content	Validation and input assistance
D16-F01 · Cycle mode	Enumeration	Number of sessions or dates or number of entitlements required.
D16-F02 · Cut-off date	Date and time	Define whether late payments are included or excluded.
D16-F03 · Entitlement status	Table by member	Subscribed, allocated, paid and remaining.
D16-F04 · Debt status	Table by member	Contributions, penalties, advances and disputes kept separate.
D16-F05 · Surplus policy	Formula	Explicit key, reserves and rounding units.
D16-F06 · Deficit policy	Decision and funding	Name who bears the loss and under which authority.
D16-F07 · Cycle carry-over	Target reference	Prohibit the same amount being carried over twice at the same time.
D16-F08 · Final validation	Minutes and controller	Require reconciliation and treatment of discrepancies.

COMMON CONTEXT AND EDITING RIGHTS

The screen always shows the association, activity or file concerned, the currency where necessary, the status and the version. The author, creation date and identifier are generated by the system. The role authorised at the workflow step determines the available actions; a hidden button does not replace the access control.

> **Summary before validation.** The screen presents the inputs, the expected result and the errors before the final action. After submission, any significant correction displays the differences; errors target the field concerned and preserve the other entered data.

Screen actions: save a draft, verify, submit, request additional information, approve or reject according to role.

C · Workflow, exceptions and transactional control

Actor	Action	Control
Manager	Stop the sessions	End condition
Treasurer	Reconcile	Cash and distributions
System	Calculate the balances	Entitlements and debts
Board or GA	Decide the allocation	Surplus and deficit
Treasurer	Settle or transfer	Linked documents
Controller	Close financially	Documented discrepancies

EXCEPTIONS AND RECOVERY

Trigger	Response	Expected trace or recovery
Last session in deficit	Keep entitlements to be served	Open funding or agree a reduction.
Payment received after closing	Allocate to the former debt	Record in an open period.
Member cannot be found	Keep the amount payable	Do not redistribute their entitlement without an applicable decision.
Unsettled advance loan	Maintain a cycle debt	Explicit transfer if the new cycle takes it over.

Trigger	Response	Expected trace or recovery
Closing cancelled	Exceptional procedure	New version and justification, no deletion.

D · Combinations, test recipes and indicators

COMBINATIONS AND QUALIFICATIONS

Function combination	Qualification	Validity condition
Closing + active credit	Conditional	Recovery continues outside the sessions.
Surplus + dividend	Conditional	Define its nature before distribution.
Default + solidarity fund	Restricted	No automatic draw on the social fund.
Renewal + change of members	Conditional	Former entitlements remain attached to their holders.
Deficit + federal advance	Compatible	Inter-entity contract and reconciliation.

TEST SCENARIOS

Case	Data or trigger	Verifiable result
D16-T01 · Twelve sessions at 600 000	Distribution 550 000 each	Receipts 7 200 000, paid 6 600 000, surplus 600 000.
D16-T02 · Cycle by entitlements	Nineteen entitlements out of twenty paid	Cycle not completed despite the indicative date being past.
D16-T03 · Defaulter already a beneficiary	Future contribution unpaid	Debt retained and guarantee enforceable per contract.
D16-T04 · Deferral of 50 000	New cycle opened	Exit from the old cycle and entry to the new, total unchanged.
D16-T05 · Surplus distributed	Three equal entitlements, 100 units	Distribution 34, 33, 33 under the published tie-break.

STEERING INDICATORS

Indicator	Formula	Reading
Unserved entitlements	Amount of entitlements still to be settled at operational closing	Do not hide them in an overall balance.
Cycle loss	Receivables written off after decision	Separate temporary default from definitively approved loss.

D17

DOMAIN 17 · OPERATIONAL SHEET

Savings, deposits, withdrawals and interest

Savings constitute a debt of the association towards the depositor where the contract provides for their return. They must not be confused with a contribution definitively acquired, a capital share exposed to loss or a promise of return. The product defines availability, interest, fees, blocking and exit treatment. The target distinguishes accounting balance, available balance and pledged balance. Interest is calculated on an explicit base and time convention; the contract states whether it is paid out, capitalised or merely attributed at the end of the period.

Manual reference: p. 29, 35, 52, 85. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Product	Demand, locked, goal-based, periodic deposit	Each product carries its own withdrawal conditions.
Payment in	Free, regular, minimum, cap	A missed payment becomes a debt only if the contract so provides.
Interest	None, simple, compound, profit participation	Do not guarantee a variable return by default.
Base	Daily balance, monthly minimum, average balance	Keep the movements and value dates used.
Withdrawal	Immediate, notice, maturity, emergency	Check liquidity and contractual entitlements separately.
Blocking	Guarantee, dispute, holder instruction	Amount, reason, duration and authority identified.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behaviour
EPA01	Available = vested balance - blocked amounts - reserved withdrawals, without counting a block twice.
EPA02	A deposit awaiting confirmation produces neither availability nor interest unless expressly agreed.
EPA03	Capitalisation is a separate movement; do not pay interest on interest that is not yet vested.
EPA04	Repayment of savings requires real liquidity and does not reduce a non-refundable contribution.

B · Form and data-entry experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced by fictitious values. Final validation runs server-side.

Functional field	Type or content	Validation and input assistance
D17-F01 · Savings product	Reference and version	Display the terms, interest and risks specific to the product.
D17-F02 · Holder	Member or representative	Check capacity and withdrawal powers.
D17-F03 · Amount deposited	Positive monetary amount	Respect currency, minimum and cap.
D17-F04 · Value date	Date	Rule determined by confirmation and channel.
D17-F05 · Rate and base	Percentage and convention	Periodicity mandatory; explicit days/year basis.
D17-F06 · Goal and term	Amount and date	Optional for demand savings; mandatory for project savings.
D17-F07 · Withdrawal notice	Duration and request	Display the earliest possible payment date.
D17-F08 · Blocking	Reservation table	Each block references a guarantee or a decision.

COMMON CONTEXT AND EDITING RIGHTS

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Screen actions: save a draft, verify, submit, request additional information, approve or reject according to role.

C · Workflow, exceptions and transactional control

Actor	Action	Control
Member	Open the product	Contract accepted
Treasurer	Confirm the deposit	Payment reconciled
System	Calculate the entitlements	Balance and interest
Member	Request the withdrawal	Notice and amount
Validator	Reserve the payment	Available balance and liquidity
Treasurer	Settle	Balance and proof

EXCEPTIONS AND RECOVERY

Trigger	Response	Expected trace or recovery
Withdrawal above the available balance	Reject the amount	Display the blocking without revealing third-party data.
Early termination	Simulate the consequences	Fees or loss of interest under the accepted contract.
Death	Freeze any release to third parties	Prepare the statement for verified rightful claimants.
Lack of liquidity	Raise a treasury alert	Do not mark the request as paid.

Trigger	Response	Expected trace or recovery
Rate changed	Create a new period	Keep the previous rates in the calculation.

D · Combinations, test recipes and indicators

COMBINATIONS AND QUALIFICATIONS

Function combination	Qualification	Validity condition
Savings + secured loan	Compatible	Atomic blocking of the pledged amount.
Savings + tontine	Compatible	Two products and two distinct entitlements.
Demand savings + illiquid investment	Conditional	Transformation limit and liquidity reserve.
Withdrawal + member leaving	Conditional	Set-offs only where authorised and explicit.
Savings + variable return	Conditional	Publication of the method and allocation decision.

TEST SCENARIOS

Case	Data or trigger	Verifiable result
D17-T01 · Balance 100 000, pledge 30 000	Withdrawal reserved 20 000	Available 50 000.
D17-T02 · Deposit 100 000 at 6 % per year	30 days on a 365-day basis	Gross interest 493,150685 before contractual rounding.
D17-T03 · Duplicate withdrawal	Same request key	A single reservation and a single payment.
D17-T04 · Capitalised interest	Following period	The new principal includes only vested interest.
D17-T05 · Unconfirmed payment in	Pending payment screen	No additional available balance.

STEERING INDICATORS

Indicator	Formula	Reading
Withdrawal coverage	Dedicated liquidity / due and reserved withdrawals	Alert if below the approved threshold.
Blocked savings	Pledged amounts / vested deposits	Monitor double pledges and expiries.

D18

DOMAIN 18 · OPERATIONAL SHEET

Loans, applications, caps and decision

Granting a loan is a risk commitment before it is a disbursement. The engine examines the member’s status, the purpose, the declared capacity, the outstanding balance, the guarantees, concentration and the availability of the fund. The cap must be the minimum of independent limits, not a sum of tolerances. The HORIZON ratio of 150 % cited in the manual is not an authorisation to create cash: it measures a portfolio cap on a basis to be specified. An accepted application reserves the funds for a limited period; the contract and the repayment schedule are accepted before payment.

Manual reference: p. 41–45, 76, 78. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Purpose	Productive, school, emergency, social, advance	Distinct products with criteria and authorised funds.
Cap	Fixed, multiple of contribution, fund ratio, capacity	Combine by minimum and deduct the relevant outstanding balance.
Term	Days, weeks, months, final date	Adapt the calendar and the calculation convention.
Disbursement	Single, tranches, payment to the supplier	Each tranche has its own conditions precedent.
Approval	Board, committee, GA, threshold delegation	Recusal of the applicant and of related parties.
Pricing	Rate, fees, optional insurance, penalty	Present the total cost and the calculation method.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behaviour
PRE01	Authorisable amount = min(free liquidity, remaining member limit, product limit, remaining concentration).
PRE02	An approved decision may expire without disbursement; the reservation is then released.
PRE03	The originator of the application cannot validate their own loan alone.
PRE04	Surety commitments and other loans are included without double counting.

B · Form and data-entry experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced by fictitious values. Final validation runs server-side.

Functional field	Type or content	Validation and input assistance
D18-F01 · Applicant and product	References	Membership and product active on the application date.
D18-F02 · Purpose and budget	Text and breakdown	Documents required depending on purpose and amount.
D18-F03 · Amount requested	Monetary	Display calculated caps and the reasons for any reduction.
D18-F04 · Term and grace period	Number and unit	The grace period states whether it covers principal only or principal and interest.
D18-F05 · Resources and charges	Periodic amounts	Dated declared data; justification according to policy.
D18-F06 · Proposed guarantees	List	Net values, availability and consent of the guarantors.
D18-F07 · Related parties	Declaration	Trigger recusal and a higher level of approval.

Functional field	Type or content	Validation and input assistance
D18-F08 · Credit offer	Contract version	Net amount received, instalments, total cost and expiry conditions.

COMMON CONTEXT AND EDITING RIGHTS

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Screen actions: save a draft, verify, submit, request additional information, approve or reject according to role.

C · Workflow, exceptions and transactional control

Actor	Action	Control
Member	Apply	Purpose and amount
Analyst	Assess	Capacity and guarantees
System	Calculate the limits	Outstanding balance and liquidity
Committee	Decide	Recusals and quorum
Member	Accept the offer	Contract and repayment schedule
Treasurer	Disburse	Conditions satisfied

EXCEPTIONS AND RECOVERY

Trigger	Response	Expected trace or recovery
Insufficient funds	Reduce or put on hold	New acceptance if the offer changes.
Guarantee refused	Return to assessment	No payment against an assumed guarantee.

Trigger	Response	Expected trace or recovery
Concurrent application	Recalculate at the time of reservation	Avoid exceeding the cumulative limit.
Offer expired	Release the reservation	New analysis before reissue.
Board member is the borrower	Apply the recusal	Keep the list of effective decision-makers.

D · Combinations, test recipes and indicators

COMBINATIONS AND QUALIFICATIONS

Function combination	Qualification	Validity condition
Loan + pledged savings	Compatible	Blocking verified before disbursement.
Loan + solidarity	Conditional	Distinguish a repayable advance from definitive aid.
Loan + death fund	Restricted	No use without authorisation under the fund policy.
Loan + collective investment	Conditional	Cumulative exposure to the same related party.
Loan in tranches + works	Compatible	Release against validated proof of progress.

TEST SCENARIOS

Case	Data or trigger	Verifiable result
D18-T01 · Liquidity 300 000, member limit 200 000	Application 250 000	Maximum authorisation 200 000 before other restrictions.
D18-T02 · Contribution 100 000, multiple of two	Retained outstanding balance 80 000	Residual limit 120 000.
D18-T03 · Two loans of 150 000	Cumulative cap 200 000	The second reservation may cover 50 000 at most.
D18-T04 · Zero-rate social loan	Term of three months	No implicit interest generated.
D18-T05 · Offer not accepted	Validity date passed	No disbursement; reservation released.

STEERING INDICATORS

Indicator	Formula	Reading
Acceptance rate	Applications approved / decisions issued	Distinguish refusal, withdrawal and incomplete file.
Borrower concentration	Outstanding balance and commitments of one member / portfolio	Also check groups of related parties.

D19

DOMAIN 19 · OPERATIONAL SHEET

Interest schedules and cost of credit

The calculation distinguishes constant amortisation, constant instalment, bullet repayment and flat interest on the initial principal. The expressions “monthly rate” and “declining interest” are not enough to determine a repayment schedule. The target fixes the basis, periodicity, day-count convention, dates, order of allocation and rounding rule. The total cost presents principal, interest and fees separately. A comparable annual rate can be calculated from the cash flows and dates using a stated method, without labelling it as a regulatory indicator until the local framework is validated.

Manual reference: p. 41–44, 54, 58. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Amortisation	Constant, constant annuity, bullet, custom	The total principal amortised equals the capital actually disbursed.
Interest	Remaining capital, initial capital, rate brackets	The contract shows the basis for each period.
Time	Contractual month, actual days/365, specified convention	No automatic conversion of a monthly rate into a simple annual rate.
Grace period	Principal only, total, with or without capitalisation	Make the interest due during the grace period visible.
Rounding	At instalment level, internal precision, final adjustment	Correct the residual on the last documented line.
Early repayment	Term reduction, instalment reduction, payoff	Recalculate only the future interest not yet accrued, as per contract.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behaviour
ECH 01	For constant amortisation, periodic principal = capital / number of instalments; interest = remaining capital before the instalment × rate.
ECH 02	For a constant annuity, A = P × i / (1 - (1 + i)^(-n)); if i = 0, A = P/n.
ECH 03	The total of the cash flows must never hide fees in the principal without a contractual explanation.
ECH 04	Any rescheduling creates a new version and keeps the before/after comparison.

B · Form and input experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced by fictitious values. Final validation runs on the server side.

Functional field	Type or content	Validation and input assistance
D19-F01 · Disbursed capital	Confirmed amount	Do not calculate on a tranche not yet paid out.
D19-F02 · Method	Enumeration	Show a simulation comparing the proposed methods.
D19-F03 · Periodic rate	Decimal and unit	Value, periodicity and effective date mandatory.
D19-F04 · Number of instalments	Positive integer	Limit according to product; specify the first date.
D19-F05 · Time convention	Documented code	Irregular instalments calculated on the selected actual dates.
D19-F06 · Fees	Table of lines	Nature, payment date, any refund and basis.
D19-F07 · Grace period	Duration and treatment	Capitalisation disabled by default unless the contract authorises it.
D19-F08 · Rounding policy	Unit and mode	Keep unrounded values and allocation differences.

COMMON CONTEXT AND EDITING RIGHTS

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> **Summary before validation.** The screen presents the inputs, the expected result and the errors before the final action. After submission, any significant correction displays the differences; errors target the field concerned and preserve the other entered data.

Screen actions: save a draft, verify, submit, request additional information, approve or reject according to role.

C · Workflow, exceptions and transactional control

Actor	Action	Control
Analyst	Enter the terms	Units and method
Engine	Calculate the lines	Principal and interest
Controller	Check the totals	Final balance zero
Member	Accept	Total cost visible
System	Freeze the version	Contract and schedule
Manager	Create an amendment if needed	Comparison retained

EXCEPTIONS AND RECOVERY

Trigger	Response	Trace or expected recovery
Zero rate	Use the dedicated branch	Avoid a division by zero in the annuity.
Last instalment with a residual	Adjust the principal	Final balance exactly zero at the holding unit.
Early payment	Simulate before validation	Reduce term or amount according to the authorised choice.
Shifted date	Recalculate per convention	Do not arbitrarily add a month of interest.

Trigger	Response	Trace or expected recovery
Fees changed after agreement	Require an amendment	Do not rewrite the accepted contract.

D · Combinations, test recipe and indicators

COMBINATIONS AND QUALIFICATIONS

Function combination	Qualification	Validity condition
Declining rate + fixed annuity	Conditional	Recalculation of the annuity or explicit variable schedule.
Grace period + capitalisation	Conditional	Explain the increase in principal and validate the framework.
Partial payment + schedule	Compatible	Allocation according to the published priority.
Currency without decimals + interest	Compatible	Internal precision then conservative final rounding.
Rescheduling + arrears	Conditional	Do not erase the late-payment history.

TEST SCENARIOS

Case	Data or trigger	Verifiable result
D19-T01 · 120 000 over three months at 3 %	Constant principal	Instalments 43 600, 42 400, 41 200; interest 7 200.
D19-T02 · Same loan with fixed interest on the initial principal	3 % of 120 000 each month	Interest 10 800; method difference explicitly visible.
D19-T03 · 100 000 in three interest-free repayments	Whole unit	Principal 33 333, 33 333, 33 334 according to the rounding chosen.
D19-T04 · Principal-only grace period	Interest due	An interest instalment appears during the grace period.

Case	Data or trigger	Verifiable result
D19-T05 · Full early repayment	No contractual penalty	Remaining capital and accrued interest only.

MANAGEMENT INDICATORS

Indicator	Formula	Reading
Contractual interest	Sum of the interest in the active schedule	Distinguish calculated, accrued and collected.
Total monetary cost	Interest and fees borne by the borrower	Show together with the net amount actually received.

D20

DOMAIN 20 · OPERATIONAL SHEET

Repayments, arrears and restructurings

A repayment allocates a confirmed payment to one or more instalments. The order of allocation — fees, penalties, interest then principal, or another authorised order — must be published. Arrears are measured on amounts fallen due and unpaid, not simply on the overall status of the loan. Days in arrears use an explicit reference date. A restructuring may change the term or the instalments without erasing the original default. Writing off is an accounting and management decision; it does not automatically mean waiving the right to recover.

Manual reference: p. 43–45, 57–58, 81. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Allocation	Chosen instalment, oldest, contractual priority	Keep the detailed breakdown of each payment.
Arrears	Grace, fixed penalty, proportional, capped	Explicit basis and periodicity; no implicit accumulation.
Reminder	Soft, formal, commission, collection	Channel, frequency and confidentiality controlled.
Restructuring	Deferral, extension, reduction, waiver	New decision and new accepted schedule.
Security	Set-off, calling the guarantee, enforcement	Trigger subject to the contract and the validated framework.
Loss	Internal provision, write-off, abandonment	Distinguish risk estimation from extinguishing the receivable.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behaviour
REC01	A payment cannot be allocated beyond its confirmed amount that has not already been allocated.
REC02	Penalties are calculated on the contractual basis, with an identifiable cap and period.
REC03	A restructuring keeps the former schedule, the arrears and the actual payments.
REC04	A recovery after write-off remains recorded and reconciled.

B · Form and input experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced by fictitious values. Final validation runs on the server side.

Functional field	Type or content	Validation and input assistance
D20-F01 · Loan and payment	References	Consistent currency and payment available.
D20-F02 · Value date	Date	Determines the cut-off of interest and arrears according to convention.
D20-F03 · Breakdown	Principal, interest, fees, penalties	Sum equal to the total allocation.
D20-F04 · Target instalment	Reference or rule	Preview the automatic order before validation.
D20-F05 · Calculated arrears	Days and basis	The grace period does not change the contractual date without an amendment.
D20-F06 · Rescheduling proposal	Before/after conditions	Show total cost and new final date.
D20-F07 · Waiver or loss	Amount and reason	Authority, cap and decision document required.

Functional field	Type or content	Validation and input assistance
D20-F08 · Collection follow-up	Action, owner, date	Prohibit public messages exposing the debt.

COMMON CONTEXT AND EDITING RIGHTS

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Screen actions: save a draft, verify, submit, request additional information, approve or reject according to role.

C · Workflow, exceptions and transactional control

Actor	Action	Control
Treasurer	Receive the payment	Confirmed proof
Engine	Allocate	Contractual order
Manager	Identify the arrears	Remaining instalments
Committee	Choose the action	Capacity and security
Member	Accept the rescheduling	Signed version
Controller	Check the position	History retained

EXCEPTIONS AND RECOVERY

Trigger	Response	Trace or expected recovery
Insufficient payment	Allocate partially	Keep the residual and its original date.
Excess payment	Create a credit or pay ahead	Explicit choice, no negative interest.

Trigger	Response	Trace or expected recovery
Penalty dispute	Isolate the disputed amount	Do not suspend the recording of the principal received.
Guarantor default	Open a separate file	Avoid unjustified multiplication of the receivable.
Write-off approved	Record without deletion	Keep the authorised recovery options.

D · Combinations, test recipe and indicators

COMBINATIONS AND QUALIFICATIONS

Function combination	Qualification	Validity condition
Repayment + dividends	Conditional	Distribute only the approved base, not the principal recovered.
Arrears + social aid	Conditional	Separate social entitlements, no automatic set-off.
Restructuring + security	Conditional	The guarantor’s consent if their commitment changes.
Loss + closure	Compatible	Keep the file off the operational balance sheet according to the validated method.
Reminder + confidentiality	Mandatory	Private messages with minimal data.

TEST SCENARIOS

Case	Data or trigger	Verifiable result
D20-T01 · Instalment 43 600, received 20 000	Priority interest of 3 600	Principal repaid 16 400, instalment residual 23 600.
D20-T02 · Double allocation	Same payment	The second attempt is refused.
D20-T03 · Three-month deferral	Loan already in arrears	Former arrears visible and new schedule versioned.
D20-T04 · Penalty capped at 5 000	Theoretical calculation 7 000	Penalty due 5 000.

Case	Data or trigger	Verifiable result
D20-T05 · Recovery after write-off	Payment of 10 000	New entry and reconciliation, never an opaque restoration.

MANAGEMENT INDICATORS

Indicator	Formula	Reading
Proposed PAR30	Remaining principal of loans more than 30 days in arrears / total outstanding	Define inclusion and date precisely; it is not merely the amount fallen due.
Recovery rate	Amounts recovered / defaulted outstanding of the cohort	Separate principal, interest and penalties.

D21

DOMAIN 21 · OPERATIONAL SHEET

Security, guarantees and commitments

The security register prevents the same savings, personal guarantee or future receivable from being pledged several times beyond its capacity. It distinguishes declared value, eligible value and reserved amount. A security is not the same as a cash receipt and does not replace the solvency assessment. A guarantor’s commitments have a cumulative cap, a duration and their own consent. Release is triggered by repayment, an approved substitution or an express decision. The procedures for enforcing an asset or a guarantee must be validated within the applicable framework before automation.

Manual reference: p. 41, 45, 78. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Collateral type	Pledged savings, personal guarantee, deposit, asset, receivable	Each type has its own evidence and restrictions.
Coverage	Full, partial, threshold, joint group	Specify the principal, interest and fees covered.
Valuation	Nominal, appraisal, prudent haircut	The eligible value does not exceed the justified value.
Reservation	Exclusive, shared and capped	Reserved amount limited to the available capacity.
Consent	Per loan, envelope, fixed term	An envelope does not authorise a silent extension.
Release	On extinction, pro rata, substitution	Check all linked commitments before returning the collateral.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behaviour
GAR01	Available capacity = eligible value - active commitments not yet released.
GAR02	A guarantor does not validate their own valuation; any conflict of interest is declared.
GAR03	Enforcement cannot produce a recovery greater than the debt and the authorised costs.
GAR04	An expired security is a coverage alert, not a deletion of the loan.

B · Form and input experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced by fictitious values. Final validation runs on the server side.

Functional field	Type or content	Validation and input assistance
D21-F01 · Guarantor or owner	Person reference	Verify identity, right over the collateral and consent.
D21-F02 · Collateral type	Type and identifier	A stable identifier allows reuse to be detected.
D21-F03 · Declared value	Amount	Valuation date and source mandatory.
D21-F04 · Haircut and eligible value	Rate and amount	Reason and valuation policy.
D21-F05 · Maximum commitment	Monetary	Sum of reservations limited to this value.
D21-F06 · Covered loans	List and order	Specify sharing and rank if several contracts.
D21-F07 · Period and conditions	Dates and text	Expiry, trigger, substitution and recourse.
D21-F08 · Consent documents	Documents	Contract version and evidence of acceptance.

COMMON CONTEXT AND EDITING RIGHTS

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Screen actions: save a draft, verify, submit, request additional information, approve or reject according to role.

C · Workflow, exceptions and transactional control

Actor	Action	Control
Borrower	Propose the collateral	Identity and proof
Guarantor	Consent	Amount and duration
Analyst	Assess	Haircut and capacity
System	Reserve	Concurrent control
Manager	Monitor	Value and expiry
Approver	Release or enforce	Condition and decision

EXCEPTIONS AND RECOVERY

Trigger	Response	Trace or expected recovery
Double pledge	Refuse the excess	Show the real available amount without disclosing the other contracts.
Withdrawal of pledged savings	Block the committed portion	Authorise only the free balance.
Impaired security	Request a top-up	Do not retroactively increase the consent.
Deceased guarantor	Review the coverage	Follow the applicable framework and the evidence.
Loan settled but fees disputed	Release according to contractual coverage	Do not retain the whole collateral without grounds.

D · Combinations, test recipe and indicators

COMBINATIONS AND QUALIFICATIONS

Function combination	Qualification	Validity condition
Personal guarantee + multiple loans	Conditional	Cumulative cap per guarantor.
Savings + urgent withdrawal	Conditional	Substitution approved before release.
Collective security + joint liability	Conditional	Distinguish contractual liability from the aid fund.
Tontine receivable + loan	Conditional	Verify assignability, date and risk of non-distribution.
Asset + loan in a different currency	Conditional	FX haircut and revaluation.

TEST SCENARIOS

Case	Data or trigger	Verifiable result
D21-T01 · Value 100 000, haircut 20 %	Existing commitment 30 000	New capacity 50 000.
D21-T02 · Two simultaneous reservations of 60 000	Capacity 100 000	It is impossible to reserve 120 000.
D21-T03 · Loan fully repaid	No other commitment	Security releasable with proof.
D21-T04 · Guarantee limited to the principal	Unpaid interest only	Do not enforce beyond the guaranteed object.
D21-T05 · Substitution of guarantor	New guarantor who has not signed	Former guarantee not released automatically.

MANAGEMENT INDICATORS

Indicator	Formula	Reading
Eligible coverage	Eligible reserved securities / covered exposure	Do not equate declared value with certain realisation.

Indicator	Formula	Reading
Securities to renew	Commitments expiring before the loans end	Alert by horizon and uncovered amount.

D22

DOMAIN 22 · OPERATIONAL SHEET

Solidarity catalogue and social entitlements

The social catalogue describes entitlements and discretionary assistance without confusing them. Each benefit defines the event, population, waiting period, seniority, required contributions, declaration deadline, documents, amount and caps. The SLD and FEV rules serve as reference configurations; they must remain customisable. A flat allowance, a per-day indemnity, a reimbursement of costs and a social loan are four different products. Combining local and national aid requires a cumulation rule and a shared event identifier where this is authorised.

Manual reference: p. 46–48, 75–76. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Nature	Contributory entitlement, exceptional assistance, social loan	The loan creates a receivable; definitive aid does not.
Event	Illness, death, birth, marriage, schooling, disaster	Extensible catalogue with its own evidence.
Amount	Flat rate, days, actual costs, brackets, percentage	Apply the minimum, the event cap and the annual cap.
Eligibility	Seniority, contribution, family link, waiting period	Assess each criterion at the selected date.
Cumulation	Prohibited, complementary, capped, independent	Specify the funds, categories and years concerned.
Exception	Emergency, force majeure, GA decision	Separate circuit with identified funding.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behaviour
SOL01	Entitlement calculated = minimum of the formula amount and the applicable residual caps.
SOL02	Approved amounts not yet paid already consume the available cap, to avoid a double commitment.
SOL03	The minimum number of days of hospitalisation and the days indemnifiable are two separate parameters.
SOL04	Local aid and national aid are not duplicates if their cumulation is explicitly authorised.

B · Form and input experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced by fictitious values. Final validation runs on the server side.

Functional field	Type or content	Validation and input assistance
D22-F01 · Benefit category	Versioned reference	Refundable or definitive nature mandatory.
D22-F02 · Covered population	Members and links	Admitted categories, age and territory where applicable.
D22-F03 · Opening conditions	List of predicates	Each refusal must provide a readable reason.
D22-F04 · Amount formula	Typed expression	Flat rate, cost or days with unit and rounding.
D22-F05 · Caps	Per event, member, household, year	Define calendar, rolling or financial year.
D22-F06 · Waiting period and deadline	Durations	Specify the start of the waiting period and the declaration deadline.
D22-F07 · Required documents	Catalogue	Minimal collection and restricted access.
D22-F08 · Cumulation and derogation	Policy	Show the order in which the caps are consumed.

COMMON CONTEXT AND EDITING RIGHTS

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Screen actions: save a draft, verify, submit, request additional information, approve or reject according to role.

C · Workflow, exceptions and transactional control

Actor	Action	Control
Board	Define the benefit	Population and budget
Social committee	Assess the criteria	Formula and fairness
Competent body	Approve	Text and funding
System	Publish the catalogue	Version and date
Member	Simulate their entitlement	Understandable reasons
Manager	Review periodically	Cost and incidents

EXCEPTIONS AND RECOVERY

Trigger	Response	Trace or expected recovery
Ambiguous condition	Block publication	Define the exact threshold and the tie case.
Event before membership	Apply the waiting period and the time rule	Do not use only the current status.
Annual cap exhausted	Refuse or open separate assistance	Do not bypass through a fictitious change of category.
Benefit withdrawn	Keep the entitlements already opened	Approved transitional policy.

Trigger	Response	Trace or expected recovery
Emergency without documents	Provisional agreement possible	Cap, regularisation deadline and reinforced control.

D · Combinations, test recipe and indicators

COMBINATIONS AND QUALIFICATIONS

Function combination	Qualification	Validity condition
Illness aid + social loan	Compatible	Two decisions and two accounting natures.
Local aid + national fund	Conditional	Cumulation and split declared.
Household cap + member cap	Conditional	The minimum of the binding residuals applies.
Waiting period + emergency	Conditional	Justified derogation without changing the general rule.
Aid + contribution debt	Conditional	Entitlement rule distinct from any set-off.

TEST SCENARIOS

Case	Data or trigger	Verifiable result
D22-T01 · FEV, four-day stay	Rate 10 000, minimum three	40 000 if all days are indemnifiable after the threshold, option to be confirmed.
D22-T02 · FEV, sixteen days	Annual cap of fifteen days unused	Maximum 150 000 depending on the configuration retained.
D22-T03 · Two approved applications	Annual cap 100 000, first one 70 000	Second limited to 30 000 before payment.
D22-T04 · Serious illness SLD	Loan product of 100 000	Create a zero-rate schedule, not definitive aid.
D22-T05 · Single event	Repeated application to the same fund	Detection and control of the duplicate.

MANAGEMENT INDICATORS

Indicator	Formula	Reading
Average cost per event	Approved aid / events covered	Separate category and exposed population.
Refusal rate by reason	Applications refused for one criterion / decisions	Identify misunderstood rules or missing documents.

D23

DOMAIN 23 · OPERATIONAL SHEET

Aid applications, payments and appeals

The aid file links the event, the person opening the entitlement, the applicant, the decision, the payment beneficiary and the fund. This separation covers payments to a family, a hospital, a supplier or several recipients. An aid decision must reserve the budget and the individual cap before disbursement. Payment may be split or provisional, with a balance subject to justification. Refusals and requests for additional documents are reasoned. An appeal is examined by an identified authority and keeps the documents of the first decision.

Manual reference: p. 46–48, 78. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Initiator	Member, representative, mandated board	Assisted entry preserves the real author.
Urgency	Normal, accelerated, provisional assistance	Target lead time and cap specific to each circuit.
Verification	Documentary, field visit, third-party confirmation	Define the authorised verifiers.
Payment	Member, beneficiary, third party, split	Sum of the destinations equal to the approved amount.
Instalments	Advance, balance, reimbursement on justification	The reservation follows the unpaid residual.
Appeal	Additional documents, review, appeal	Authority and deadlines stated in the notification.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behaviour
AID01	The complete file is assessed with the benefit version applicable to the event or to the contractual date retained.
AID02	The decision, the reservation and the cap tracking are atomic.
AID03	The treasurer receives the information needed for payment, without automatic access to medical documents.
AID04	Recovery of an undue payment creates a separate receivable, with reason and appeal.

B · Form and input experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced by fictitious values. Final validation runs on the server side.

Functional field	Type or content	Validation and input assistance
D23-F01 · Event	Type, date, person	Search for linked files before creation.
D23-F02 · Applicant	Reference and capacity	Verify the mandate if acting for someone else.
D23-F03 · Documents and verification	Documents and report	Complete status, additional documents requested or derogation.
D23-F04 · Entitlement calculation	Formula detail	Show consumed caps and the proposed amount.
D23-F05 · Decision	Agreement, refusal, additional documents	Reason, signatories and abstentions.
D23-F06 · Payment split	Recipient and amount	Sum equal to the agreement, verified contact details.
D23-F07 · Instalment schedule	Dates and conditions	Reservation of the balance and possible expiry.
D23-F08 · Appeal and regularisation	Linked file	Keep the new evidence without overwriting the old.

COMMON CONTEXT AND EDITING RIGHTS

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Screen actions: save a draft, verify, submit, request additional information, approve or reject according to role.

C · Workflow, exceptions and transactional control

Actor	Action	Control
Applicant	Declare the event	Date and documents
Social manager	Verify	Eligibility and duplicate
Committee	Decide	Amount and reason
System	Reserve	Cap and fund
Treasurer	Pay the recipients	Individual proof
Controller	Close the file	Balance and supporting documents

EXCEPTIONS AND RECOVERY

Trigger	Response	Trace or expected recovery
Missing document	Request additional documents	Show what is missing and the deadline.
Insufficient funds	Place on a funded waiting list or escalate	Do not suggest that the agreement equals immediate payment.
Wrong recipient	Suspend the instalment	Independent verification before resuming.
Amount reduced on appeal	Correct reservations and the debt	Handle any undue payment separately.

Trigger	Response	Trace or expected recovery
Partially failed disbursement	Keep the residual	Do not replay instalments already paid.

D · Combinations, test recipe and indicators

COMBINATIONS AND QUALIFICATIONS

Function combination	Qualification	Validity condition
Aid + third-party payment	Compatible	Proof of payment and economic beneficiary kept separate.
Emergency + dual validation	Conditional	Pre-established emergency delegation, reviewed afterwards.
Two funds + one event	Conditional	Split of funding without double payment.
Appeal + following year	Conditional	Cap attached to the period defined by contract.
Undue payment + contribution	Distinct	Two receivables and two reasons, set-off allowed only.

TEST SCENARIOS

Case	Data or trigger	Verifiable result
D23-T01 · FEV death 300 000	Split 200 000, 50 000, 50 000	Three payments, total exactly 300 000.
D23-T02 · Agreement 70 000, advance 30 000	Balance not paid	Remaining reservation 40 000 and status partially paid.
D23-T03 · Two concurrent committees	Same residual cap	No double consumption beyond the cap.
D23-T04 · Successful appeal	Initial refusal archived	New linked decision and explanation retained.
D23-T05 · Payment to a hospital	Private medical document	The receipt does not reveal the diagnosis.

MANAGEMENT INDICATORS

Indicator	Formula	Reading
Payment lead time	First payment minus complete file	Separate the processing time from the treasury time.
Aid committed but unpaid	Agreements minus payments and cancellations	Measure of the social commitments to be funded.

D24

DOMAIN 24 · OPERATIONAL SHEET

Solidarity reserves and liquidity forecasting

The sustainability of a social fund depends on contributions, the frequency of events, the cost of aid and the payment calendar. The target provides a prospective budget and stress simulations, without treating the fund as an approved insurance product. Commitments already approved are separated from probable events and from mere applications. A safety reserve protects urgent aid; an exceptional call allows a shock to be handled according to an adopted rule. The forecast must state its assumptions and present several trajectories, without promising coverage of unknown events.

Manual reference: p. 46–48, 52, 55, 75–76. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Funding	Regular contribution, exceptional call, donation, endowment	Allocate each resource and distinguish refundable from acquired.
Reserve	Fixed amount, months of expenses, percentage	The threshold is a management policy, not an absolute guarantee.
Forecast	Monthly, annual, rolling horizon	Update from actual collections.
Events	Historical frequency, assumption, shock	Do not infer an individual risk from private medical data.
Corrective measure	Contribution revision, capped aid, funding	Entitlements already acquired require transitional treatment.
Scenario	Central, low collection, simultaneous events	Compare the final cash position and the minimum reached during the year.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behaviour
RES01	Projected balance = opening balance + confirmed or expected inflows - expected outflows, by date.
RES02	A contribution receivable is not a certain inflow; apply an explicit collection assumption.
RES03	Approved aid items are identified commitments and are not counted again in the future frequency.
RES04	Measures reducing entitlements follow a decision and an effective date, never a hidden adjustment.

B · Form and input experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced by fictitious values. Final validation runs on the server side.

Functional field	Type or content	Validation and input assistance
D24-F01 · Forecast period	Dates	Horizon and frequency of the calculation points.
D24-F02 · Opening balance	Reconciled amount	Reference date and accounts included.
D24-F03 · Exposed population	Headcount by category	Persons covered distinct from billed memberships.
D24-F04 · Expected contributions	Rate and calendar	Collection assumption separate from the call rate.
D24-F05 · Event frequencies	Assumptions by category	Source and period; a zero observed does not mean zero risk.
D24-F06 · Cost per event	Amount or distribution	Use the caps and rules of the catalogue.
D24-F07 · Target reserve	Formula	Specify whether it is ring-fenced or merely monitored.
D24-F08 · Corrective actions	Decisions and triggers	Owner, date and impact on entitlements.

COMMON CONTEXT AND EDITING RIGHTS

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C · Workflow, exceptions and transactional control

Actor	Action	Control
Treasurer	Reconcile the opening	Actual balances
Social manager	Estimate the needs	Events and commitments
Engine	Project	Cash-flow calendar
Board	Test the shocks	Deficit and reserve
GA or authority	Decide the measures	Entitlements and funding
Controller	Compare with the actuals	Variances explained

EXCEPTIONS AND RECOVERY

Trigger	Response	Trace or expected recovery
History too short	Use prudent assumptions	Show uncertainty and sensitivity.
Simultaneous shock	Trigger the emergency plan	Approved priority and funding.
Calls not collected	Revise the forecast	Do not maintain a theoretical liquidity.
Earmarked donation	Restrict its use	The donation does not fund all aid indiscriminately.

Trigger	Response	Trace or expected recovery
Insufficient reserve	Escalate before an unfunded payment	Document the trade-offs and deadlines.

D · Combinations, test recipe and indicators

COMBINATIONS AND QUALIFICATIONS

Function combination	Qualification	Validity condition
Solidarity + investment	Conditional	Horizon and liquidity compatible with the entitlements due.
Exceptional call + household	Conditional	Explicit basis and exemptions.
National reserve + section	Conditional	Support agreement and possible repayment.
Shock + capped aid	Conditional	Preserve acquired entitlements and justify the trade-offs.
Earmarked donation + general deficit	Restricted	Respect the purpose of the allocation.

TEST SCENARIOS

Case	Data or trigger	Verifiable result
D24-T01 · 100 members at 6 000 per month	90 % collection over twelve months	Expected inflows 6 480 000, not 7 200 000.
D24-T02 · Three deaths at 900 000	Same month	Outflows 2 700 000 before other benefits.
D24-T03 · Aid of 70 000 already approved	Present in the portfolio	Do not add it a second time as a future claim.
D24-T04 · Positive annual final balance	Deficit in March	The monthly simulation must flag the funding need.

Case	Data or trigger	Verifiable result
D24-T05 · Dedicated reserve	Productive loan requested	Refusal if it would breach the locked threshold.

MANAGEMENT INDICATORS

Indicator	Formula	Reading
Months of coverage	Dedicated liquidity / monthly average of relevant outflows	Show the seasonality risk.
Forecast vs actual variance	Actual flows minus forecast flows	Break down collection, frequency and unit cost.

D25

DOMAIN 25 · OPERATIONAL SHEET

Capital investment and dividends

The investment fund tracks contributions, shares, projects, income, losses and distribution decisions. Contributions are not profits. The target allows a capital allocation key at a given date, with constant shares, with capital weighted by duration, or with an approved combination. The manual uses, for certain interest calculations, the capital at the loan date; this rule must not be silently replaced by holdings at the repayment date. A distributable result and available cash are two distinct constraints.

Manual reference: p. 43–44, 53–58, 73–76. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Contribution	Capital, repayable advance, gift, reinvested result	Qualify each entry before allocating shares.
Participation	Fixed shares, nominal value, capital units	Keep a history of issues, transfers and buybacks.
Income	Interest, project return, discount, disposal	Distinguish gross, charges and net result.
Key	Shares, fixed capital, capital-days, hybrid key	The key and the period are voted before the calculation.
Allocation	Reserve, distribution, carry-forward, reinvestment	The sum of the allocations equals the available result.
Payment	Immediate, instalments, credit to account	Do not mark a dividend as paid merely on publication.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behaviour
DIV01	Share = member weight / sum of weights; a zero sum forbids the calculation.
DIV02	The sum of the allocated amounts equals exactly the decided amount, after handling remainders.
DIV03	The principal repaid on a loan is not part of distributable interest.
DIV04	The distribution run retains base, key, population, rounding, decision and payments.

B · Form and data-entry experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced by fictitious values. Final validation runs server-side.

Functional field	Type or content	Validation and data-entry assistance
D25-F01 · Financial year and run	Unique identifier	A published version cannot be overwritten.
D25-F02 · Opening result	Justified amount	Reconciliation with retained income and charges.
D25-F03 · Reserves and carry-forwards	Decided amounts	Reduce the distributable base per decision.
D25-F04 · Allocation key	Versioned method	Capital at a date or capital weighted over a period.
D25-F05 · Population and weights	Table by member	Retain entries, exits and transfers.
D25-F06 · Rounding	Unit and tie-break	Largest-remainder method or another approved method.
D25-F07 · Payment availability	Cash amount	Define due dates if the distribution is not immediately payable.

Functional field	Type or content	Validation and data-entry assistance
D25-F08 · Approval and publication	Minutes	Show the simulation, then the final state.

COMMON CONTEXT AND EDITING RIGHTS

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Screen actions: save a draft, verify, submit, request additional information, approve or reject according to role.

C · Workflow, exceptions and transactional control

Actor	Action	Control
Accountant	Establish the base	Income and charges
System	Compute the weights	Capital history
Controller	Check the rounding	Sum preserved
GA	Allocate the result	Formal decision
Treasurer	Pay or reinvest	Choice and evidence
Auditor	Replay the run	Immutable inputs

EXCEPTIONS AND RECOVERY

Trigger	Response	Expected trace or recovery
Total capital nil	Block the distribution	Request another approved key.
Member exited	Apply historical rights	Do not delete their vested share.

Trigger	Response	Expected trace or recovery
Positive result without liquidity	Publish a payable debt	Authorised and transparent schedule.
Error after publication	Cancel and replace the run	Reverse the effects already accounted for.
Project loss	Reduce the result per validated method	Do not promise a guaranteed return.

D · Combinations, test recipes and indicators

COMBINATIONS AND QUALIFICATIONS

Function combination	Qualification	Validity condition
Investment + demand savings	Conditional	Do not confuse risk capital with a repayable deposit.
Dividend + member debt	Conditional	Explicit set-off and detailed statement.
Capital-days + transfer	Compatible	Allocate the weight between holders according to dates.
Tontine discount + dividend	Conditional	Nature and allocation decided beforehand.
Reinvestment + vote	Conditional	The payment decision does not automatically create political rights.

TEST SCENARIOS

Case	Data or trigger	Verifiable result
D25-T01 · 5 433 499 allocated across seven shares	Whole monetary unit	One share 776 215 and six shares 776 214 depending on the tie-break.
D25-T02 · Interest 5 219 831, base 5 433 499	HORIZON figures from the manual	Difference 213 668 to be justified, without inventing an origin.
D25-T03 · Contribution during the year	Capital-days key	Only the effective duration contributes to the weight.

Case	Data or trigger	Verifiable result
D25-T04 · Principal repayment 100 000	No interest	Dividend base unchanged.
D25-T05 · Two validations of the run	Same financial year and version	Only one dividend debt is created.

STEERING INDICATORS

Indicator	Formula	Reading
Realised return	Attributable net result / retained average capital	Specify period, losses and denominator method.
Dividends payable	Published amounts less settlements and reinvestments	Distinguish debt from available cash.

D26

DOMAIN 26 · OPERATIONAL SHEET

Projects, grants, donations and campaigns

The ARP must track collective projects beyond the cash box: objectives, beneficiaries, budget, schedule, tasks, deliverables and evidence of fund use. A grant or an earmarked donation carries conditions that may prohibit certain expenses. The fundraising campaign distinguishes pledge from receipt, unrestricted from earmarked amount, member contributor from external donor. Multi-year projects keep their total budget and their execution by financial year. A report to the funder reconciles amounts received, justified expenses, unjustified advances and the balance to be returned.

Manual reference: p. 7, 29, 35, 66. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Project	Infrastructure, education, social action, event, economic activity	Define expected results and owner.
Resource	Special contribution, donation, grant, loan, in-kind contribution	Qualify repayment rights and allocation.
Budget	Line items, lots, financial years, revisions	Do not confuse awarded budget with funds received.
Collection	Target, tier, counterpart, anonymous to the public	Accounting identity may still be required under restricted access.
Execution	Internal, contractor, volunteering, mixed	Link tasks, purchases and acceptance of deliverables.
Report	Financial, narrative, indicators, supporting documents	Dated version validated before transmission.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behaviour
PRO01	A donation pledge does not finance an outgoing payment until it is received or secured by authorised funding.
PRO02	An expense may be split across projects, but its total amount is counted only once.
PRO03	A funder's restrictions are tested before commitment, not only in the final report.
PRO04	Project closure handles advances, remaining balances, assets and reporting obligations.

B · Form and data-entry experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced by fictitious values. Final validation runs server-side.

Functional field	Type or content	Validation and data-entry assistance
D26-F01 · Project sheet	Title, purpose, territory	Describe the concrete result and the target population.
D26-F02 · Owner and team	References	Mandate and validation responsibilities.
D26-F03 · Detailed budget	Costed lines	Quantity, unit cost, funding and period.
D26-F04 · Pledged resources	Amounts and due dates	Statuses: pledged, contracted, received.
D26-F05 · Funding conditions	Document and rules	Eligible expenses, co-financing and restitution.
D26-F06 · Milestones	Dates and deliverables	Verifiable acceptance criteria.
D26-F07 · Expenses and evidence	Linked documents	Analytical imputation and approval.
D26-F08 · Closure	Report and allocation of the balance	Handle remaining balances per agreement.

COMMON CONTEXT AND EDITING RIGHTS

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C · Workflow, exceptions and transactional control

Actor	Action	Control
Project lead	Propose the project	Objective and budget
Committee	Evaluate	Usefulness and funding
Competent body	Authorise	Budget and limits
Team	Execute	Milestones and purchases
Controller	Verify	Documents and results
Funder or board	Close	Report and remaining balance

EXCEPTIONS AND RECOVERY

Trigger	Response	Expected trace or recovery
Collection below target	Apply the announced scenario	Reduce, extend or refund per conditions.
Ineligible expense	Block imputation to the funder	Seek separate authorised funding.
Construction delay	Revise the milestones	Budget impact and decision retained.
Unjustified advance	Keep an advance account	No definitive expense without the required evidence.

Trigger	Response	Expected trace or recovery
In-kind donation	Value separately	Do not increase bank cash.

D · Combinations, test recipes and indicators

COMBINATIONS AND QUALIFICATIONS

Function combination	Qualification	Validity condition
Project + group purchasing	Compatible	Distinguish project goods from individual orders.
Donation + solidarity	Compatible	Allocation to a category or to the unrestricted fund.
Grant + loan	Conditional	The loan cannot use a resource prohibited for that use.
Project + volunteering	Compatible	Time valuation separate from cash outflows.
Campaign + diaspora	Conditional	Currency, fees, identity and channel restrictions.

TEST SCENARIOS

Case	Data or trigger	Verifiable result
D26-T01 · Pledge 1 000 000, received 400 000	Project with no other funds	Available at most 400 000 before commitments.
D26-T02 · Invoice 100 000 split 60/40	Two projects	Total expense remains 100 000.
D26-T03 · Grant earmarked for supplies	Request for refreshments	Refused on this funding.
D26-T04 · Project completed with 50 000	Restitution agreement	Funder receivable or documented refund.
D26-T05 · Milestone not accepted	Conditional tranche	Disbursement blocked.

STEERING INDICATORS

Indicator	Formula	Reading
Budget execution	Validated expenses / revised budget	Also show unpaid commitments.
Achievement of results	Accepted deliverables / planned deliverables	Do not equate high spending with achieved impact.

D27

DOMAIN 27 · OPERATIONAL SHEET

Group purchasing, inventory and in-kind contributions

Group purchasing links member requests, supplier order, advances, receipt, distribution and final settlement. Physical quantities and monetary values are tracked separately. An in-kind contribution may settle an obligation if an approved valuation rule allows it; it does not feed a tontine's cash pot until it is sold. Units, qualities, lots, losses, transport and fees must be explicit. Distribution may follow ordered quantities, shares or an approved priority in case of shortage.

Manual reference: p. 29, 35, 72. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Order	Pre-order, collective stock, purchase by lot	Define who bears unsold items and cancellations.
Unit	Piece, kilogram, bag, litre, hour	Versioned conversions; the weight of a bag must be known.
Quality	Grade, inspection, rejection, markdown	Retain the decision at receipt.
Cost	Price, transport, taxes and applicable fees	Allocate by quantity, weight or value per rule.
Distribution	Proportional, exact order, priority	Handle indivisible units and remainders.
Stock	Inbound, outbound, transfer, loss, count	Each movement carries lot, location and supporting document.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behaviour
ACH01	Final stock = opening stock + inbound - outbound - validated losses, in a consistent unit.
ACH02	The sum of the imputed costs equals the retained collective cost after rounding.
ACH03	A partial receipt does not mark the entire order as delivered.
ACH04	In-kind valuation must not be counted both as a monetary payment and as an additional donation.

B · Form and data-entry experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced by fictitious values. Final validation runs server-side.

Functional field	Type or content	Validation and data-entry assistance
D27-F01 · Item and unit	Catalogue reference	Base unit and approved conversions.
D27-F02 · Requested quantity	Positive number	Respect minimum, lot and delivery capacity.
D27-F03 · Supplier and quotation	References	Retain the selection and any conflicts of interest.
D27-F04 · Price and fees	Monetary lines	Currency and allocation method.
D27-F05 · Member advance	Linked payment	An unallocated advance does not equal delivery.
D27-F06 · Receipt	Quantities and quality	Accepted, rejected and missing kept separate.
D27-F07 · Distribution	Member, lot, quantity	Signature or proof of handover.
D27-F08 · Regularisation	Balance and refund	Close price, quantity and advance discrepancies.

COMMON CONTEXT AND EDITING RIGHTS

The screen always shows the association, activity or file concerned, the currency where necessary, the status and the version. The author, creation date and identifier are generated by the system. The role authorised at the workflow step determines the available actions; a hidden button does not replace the access control.

> **Summary before validation.** The screen presents the inputs, the expected result and the errors before the final action. After submission, any significant correction displays the differences; errors target the field concerned and preserve the other entered data.

Screen actions: save a draft, verify, submit, request additional information, approve or reject according to role.

C · Workflow, exceptions and transactional control

Actor	Action	Control
Members	Order	Confirmed quantities
Buyer	Compare offers	Price and quality
Validator	Commit	Budget and selection
Storekeeper	Receive	Quantities and inspection
Manager	Distribute	Rule and costs
Treasurer	Regularise	Balances and refunds

EXCEPTIONS AND RECOVERY

Trigger	Response	Expected trace or recovery
Insufficient delivery	Distribute per rule	Inform members and refund those not delivered.
Non-compliant product	Quarantine	Return or approved markdown.
Stock loss	Declare an incident	Cross-checked count and allocation of the loss.
Missing conversion	Block the equivalence	Do not assume the weight of a bag.

Trigger	Response	Expected trace or recovery
Higher final price	Request the required agreement	Do not automatically debit beyond the accepted cap.

D · Combinations, test recipes and indicators

COMBINATIONS AND QUALIFICATIONS

Function combination	Qualification	Validity condition
In-kind + monetary tontine	Conditional	Sale or funding before cash distribution.
Purchasing + contribution	Conditional	Advance allocation distinct from the entry fee.
Stock + project	Compatible	Imputation to the project at the time defined by the validated method.
Shortage + unequal shares	Conditional	Rounding and tie-break published.
Supplier return + closed cash box	Compatible	Credit note and movement in an open period.

TEST SCENARIOS

Case	Data or trigger	Verifiable result
D27-T01 · 100 kg received, 90 handed over, 2 lost	Opening stock nil	Final stock 8 kg.
D27-T02 · Fees 10 000 for 100 kg	Allocation by weight	100 per kg added to the cost.
D27-T03 · Contribution of 20 kg at 500	Approved valuation	Value 10 000; cash unchanged.
D27-T04 · Order 50 units, receipt 30	Partial delivery	20 units remain to be received.

Case	Data or trigger	Verifiable result
D27-T05 · Two members for the last unit	Concurrent validation	Only one stock issue possible.

STEERING INDICATORS

Indicator	Formula	Reading
Delivery rate	Accepted quantities / ordered quantities	Measure per homogeneous unit.
Inventory discrepancy	Physical stock - theoretical stock	Value separately from price variances.

D28

DOMAIN 28 · OPERATIONAL SHEET

Physical assets, volunteering and time

A complete ARP tracks non-financial resources: buildings, equipment, vehicles, rooms, skills and volunteer time. The asset register states ownership, allocation, condition, holder and maintenance. Lending equipment is not a transfer of ownership. The time bank measures service units and can operate without monetary conversion. If a valuation is published in a report, it must be separated from receipts. Reservations prevent the same asset or the same person from being assigned to two incompatible missions.

Manual reference: p. 7, 35, 66. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Asset	Durable, consumable, rented, lent by a third party	Distinguish ownership from mere custody.
Allocation	Association, section, project, member custodian	Explicit owner and period.
Reservation	Slot, duration, priority, deposit	Prevent overlaps and allow for late returns.
Maintenance	Preventive, corrective, periodic inspection	An unavailable asset cannot be reserved normally.
Volunteering	Skill, mission, hours, validation	The beneficiary validates the service performed.
Valuation	None, indicative rate, time credit	Do not create implicit remuneration.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behaviour
PAT01	An asset has a location and a principal holder at a given time.
PAT02	Validated hours cannot be duplicated between two missions in the same slot without justification.
PAT03	A time credit does not automatically become a monetary receivable.
PAT04	A disposal of an asset requires a decision, a price or a justified free transfer, and removal from the register.

B · Form and data-entry experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced by fictitious values. Final validation runs server-side.

Functional field	Type or content	Validation and data-entry assistance
D28-F01 · Resource	Asset or skill	Stable identifier and category.
D28-F02 · Owner	Organisation or third party	Proof of ownership or availability agreement.
D28-F03 · Location and holder	References	Transfer history and acknowledgement of receipt.
D28-F04 · Value and date	Amount and justification	Inventory method distinct from the possible sale price.
D28-F05 · Condition and maintenance	Status and due dates	Block use if danger or failure is reported.
D28-F06 · Reservation or mission	Dates and owner	Conflicts and capacity verified.
D28-F07 · Time performed	Hours and validation	Distinguish estimated, declared and accepted.
D28-F08 · Return or closure	Condition, discrepancies, signature	Handle breakage, delay and missing consumables.

COMMON CONTEXT AND EDITING RIGHTS

The screen always shows the association, activity or file concerned, the currency where necessary, the status and the version. The author, creation date and identifier are generated by the system. The role authorised at the workflow step determines the available actions; a hidden button does not replace the access control.

> **Summary before validation.** The screen presents the inputs, the expected result and the errors before the final action. After submission, any significant correction displays the differences; errors target the field concerned and preserve the other entered data.

Screen actions: save a draft, verify, submit, request additional information, approve or reject according to role.

C · Workflow, exceptions and transactional control

Actor	Action	Control
Manager	Register	Identity and ownership
Requester	Reserve	Item and dates
Supervisor	Authorise	Availability and priority
Custodian	Hand over	Initial condition
User	Return or declare	Time and final condition
Controller	Close	Discrepancies and maintenance

EXCEPTIONS AND RECOVERY

Trigger	Response	Expected trace or recovery
Double booking	Refuse the overlap	Offer another slot.
Damaged equipment	Freeze availability	Cross-checked report and repair.
Volunteer absent	Reassign the mission	No time credit for work not performed.
Hours disputed	Request validation	Do not publish an unaccepted total as performed.
Asset lost	Open an incident	Disposal decision and liability handled separately.

D · Combinations, test recipes and indicators

COMBINATIONS AND QUALIFICATIONS

Function combination	Qualification	Validity condition
Volunteering + contribution	Conditional	Accepted equivalence and cap, without fictitious cash.
Equipment + project	Compatible	Allocation and usage cost distinct.
Room + event	Compatible	Reservation linked to the event capacity.
Time credit + financial loan	Distinct	No monetary guarantee assumed.
Disposal + related party	Conditional	Recusal, valuation and decision documented.

TEST SCENARIOS

Case	Data or trigger	Verifiable result
D28-T01 · Room booked from 10:00 to 12:00	New request at 11:00	Conflict detected.
D28-T02 · Four-hour mission	Three hours validated	Time credit equal to three.
D28-T03 · Equipment donation of 100 000	No sale	Assets increased; cash unchanged.
D28-T04 · Equipment loan	Returned without damage	No transfer of ownership.
D28-T05 · Overdue blocking maintenance	Reservation requested	Refusal, or safety waiver not authorised per policy.

STEERING INDICATORS

Indicator	Formula	Reading
Equipment availability	Usable time / planned service time	Deduct maintenance and breakdowns.
Validated volunteer hours	Sum of accepted hours per period	Present any indicative valuation separately.

D29

DOMAIN 29 · OPERATIONAL SHEET

Events, refreshments and logistics

The event covers festive gatherings, ceremonies, training, cultural events and field actions. It manages registration, capacity, invitations, services, transport, accommodation and budget. Refreshments are a specific by-product: contribution per session and flat fee to the host. The SLD manual mentions 3 000 FCFA per member and a 150 000 FCFA flat fee; with fewer than fifty paid contributions, the flat fee is not funded by the session alone. The software must therefore display carry-forward, subsidy or deficit rather than pay out of a fictitious cash box.

Manual reference: p. 37, 49–51, 64. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Access	Members, families, guests, public	Price categories and minimum information.
Capacity	Overall, per room, per service	Reservation and waiting list.
Price	Free, flat fee, free contribution, tiers	Refund conditions displayed.
Refreshments	Fixed contribution, actual cost, host flat fee	Specify the base and the funding of the flat fee.
Set-off	Arrears of the same fund, other authorised debts	The withholding detail is visible before payment.
Logistics	Meals, transport, equipment, reception	Services booked and delivered separately.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behaviour
EVE01	A confirmed registration does not exceed available capacity unless overbooking is explicitly authorised.
EVE02	Host net flat fee = approved flat fee - authorised withholdings, with no hidden negative amount.
EVE03	A withholding for arrears reduces a debt; it is not a new external receipt.
EVE04	A cancellation handles tickets, non-recoverable expenses and already committed flat fees separately.

B · Form and data-entry experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced by fictitious values. Final validation runs server-side.

Functional field	Type or content	Validation and data-entry assistance
D29-F01 · Event	Type, title, owner	Objective and expected population.
D29-F02 · Dates and venues	Slots	Check room and equipment reservations.
D29-F03 · Capacity	Integers per service	Seats reserved, confirmed and pending.
D29-F04 · Prices and exemptions	Amounts	Rule by category and funding of free places.
D29-F05 · Registration	Participant and options	Separate consents for photo and communication.
D29-F06 · Host and flat fee	Member, amount, fund	Check eligibility and cash availability.
D29-F07 · Authorised withholdings	Itemised debts	Agreement or statutory basis and cap.
D29-F08 · Budget and closure	Receipts, expenses, documents	Compare forecast, committed and actual.

COMMON CONTEXT AND EDITING RIGHTS

The screen always shows the association, activity or file concerned, the currency where necessary, the status and the version. The author, creation date and identifier are generated by the system. The role authorised at the workflow step determines the available actions; a hidden button does not replace the access control.

> **Summary before validation.** The screen presents the inputs, the expected result and the errors before the final action. After submission, any significant correction displays the differences; errors target the field concerned and preserve the other entered data.

Screen actions: save a draft, verify, submit, request additional information, approve or reject according to role.

C · Workflow, exceptions and transactional control

Actor	Action	Control
Organiser	Plan	Budget and capacity
Board	Authorise	Funding
Participants	Register	Seats and payment
Team	Welcome	Attendance and services
Treasurer	Settle the charges	Flat fee and withholdings
Organiser	Close	Review and incidents

EXCEPTIONS AND RECOVERY

Trigger	Response	Expected trace or recovery
Capacity reached	Open a waiting list	Do not take a firm fee without a confirmed seat.
Participant cancellation	Apply the conditions	Calculate the refund and the justified fees.
Insufficient receipts	Request funding or revise	No implicit negative cash balance.
Arrears greater than the flat fee	Limit the net to zero	Keep the remaining debt separately.

Trigger	Response	Expected trace or recovery
Event postponed	Offer deferral or refund	Retain the participant's decision.

D · Combinations, test recipes and indicators

COMBINATIONS AND QUALIFICATIONS

Function combination	Qualification	Validity condition
Refreshments + tontine	Compatible	Separate cash boxes and obligations.
Event + subsidised project	Conditional	Eligible expenses to be verified.
Guest + social aid	Distinct	A registration does not create membership or social rights.
Set-off + loan	Conditional	Explicit authorisation, statement and limit.
Photo + directory	Conditional	Publication choice and controlled access.

TEST SCENARIOS

Case	Data or trigger	Verifiable result
D29-T01 · 40 members pay 3 000	Host flat fee 150 000	Receipts 120 000, additional need 30 000.
D29-T02 · Flat fee 150 000, arrears 12 000	Authorised withholding	Net payable 138 000 and debt reduced by 12 000.
D29-T03 · Two registrations for the last seat	Concurrent transactions	Only one confirmation.
D29-T04 · Free place granted	Price 5 000	No fictitious payment of 5 000.
D29-T05 · Event postponed	Ticket already paid	Payment retained or refunded, never duplicated.

STEERING INDICATORS

Indicator	Formula	Reading
Event coverage	Earned receipts / committed charges	Pledges are not earned receipts.
Attendance rate	Participants present / confirmed registrations	Distinguish cancellations and no-shows.

D30

DOMAIN 30 · OPERATIONAL SHEET

Community, services, training and awards

The community space complements management operations with a directory, messages, announcements, mutual aid, a marketplace, training and awards. Their presence in the manual's navigation does not prove complete business coverage; the capabilities below are extensions to be specified and validated. Informal exchanges cannot approve a loan or amend the rules without going through the official process. Artificial-intelligence-assisted advice remains proposals that are reviewed, never automatic financial decisions. Honorary awards require a register, criteria and a decision, like the EPI mentioned among the manual's limitations.

Manual reference: p. 26–27, 60–66, 82. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Communication	Official announcement, group, private message, support	Separate channels and permissions.
Directory	Private, visible to members, voluntary publication	Contact details masked according to choice and role.
Marketplace	Listing, service request, supervised order	Define liability, payment and dispute handling.
Training	Pathway, quiz, attendance, certificate	A certificate states the criteria actually met.
Award	Nomination, review, jury, award	Documented criteria and conflicts of interest.
AI assistance	Search, explanation, draft, simulation	Cite the rules and require human confirmation of actions.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behaviour
COM01	A group message does not constitute a statutory decision without an explicit adoption procedure.
COM02	Contact details and financial difficulties are not exposed by default in the directory.
COM03	Gamification must not reveal members' debts or social files.
COM04	AI assistance receives only authorised data and cannot bypass the requester's rights.

B · Form and data-entry experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced by fictitious values. Final validation runs server-side.

Functional field	Type or content	Validation and data-entry assistance
D30-F01 · Space and audience	Group or channel	Check membership and visibility before publication.
D30-F02 · Content	Text, media, document	Moderation and file control.
D30-F03 · Listing or service	Category, price, conditions	Specify whether payment is integrated or a mere connection.
D30-F04 · Report	Reason and evidence	Complainant confidentiality per policy.
D30-F05 · Training	Objectives and modules	Pass criteria and content version.
D30-F06 · Award	Candidate, criteria, jury	Documents and recusals of related persons.
D30-F07 · Publication consent	Dated choice	Name, photo and contact details separate.
D30-F08 · Assisted action	Proposal and confirmation	The summary shows effects and amounts before execution.

COMMON CONTEXT AND EDITING RIGHTS

The screen always shows the association, activity or file concerned, the currency where necessary, the status and the version. The author, creation date and identifier are generated by the system. The role authorised at the workflow step determines the available actions; a hidden button does not replace the access control.

> **Summary before validation.** The screen presents the inputs, the expected result and the errors before the final action. After submission, any significant correction displays the differences; errors target the field concerned and preserve the other entered data.

Screen actions: save a draft, verify, submit, request additional information, approve or reject according to role.

C · Workflow, exceptions and transactional control

Actor	Action	Control
Member	Create or request	Audience and content
System	Control access	Scope and files
Moderator or jury	Review	Criteria and reports
Competent authority	Publish or award	Reasoned decision
Participants	Interact or learn	Service rules
Supervisor	Handle appeals	History and removal

EXCEPTIONS AND RECOVERY

Trigger	Response	Expected trace or recovery
Content reported	Hide if necessary	Reasoned review and appeal.
Suspected fraudulent listing	Suspend the listing	Do not characterise definitively without investigation.
Erroneous AI advice	Correct and trace	No automatic write to reverse if confirmation is absent.
Award contested	Re-examine the file	Retain criteria and deliberation.

Trigger	Response	Expected trace or recovery
Departing member	Remove private access	Retain contributions per applicable policy.

D · Combinations, test recipes and indicators

COMBINATIONS AND QUALIFICATIONS

Function combination	Qualification	Validity condition
Marketplace + contribution	Distinct	Commercial purchase and membership obligation separate.
Chat + vote	Conditional	Refer to the formal ballot and its electoral roll.
Training + treasurer role	Compatible	Training prerequisite possible before authorisation.
AI + medical file	Restricted	No transmission without a suitable purpose and authorisation.
Gamification + solidarity	Restricted	Avoid named rankings based on aid received.

TEST SCENARIOS

Case	Data or trigger	Verifiable result
D30-T01 · Member of another association	Link to a private group	Access denied server-side.
D30-T02 · AI asked to approve a loan	No decision mandate	Produces an explanation or a draft, not an approval.
D30-T03 · Directory exported	Member hid their phone number	Phone number absent from the export.
D30-T04 · Quiz completed without reaching the threshold	Certificate of achievement	Not issued; participation may be certified separately.
D30-T05 · EPI award	Jury and criteria recorded	Decision, date and evidence viewable according to rights.

STEERING INDICATORS

Indicator	Formula	Reading
Requests resolved	Support cases closed with a reason / cases received	Separate resolution from mere closure.
Training validated	Members who met the criteria / enrolled	State the pathway version and the period.

D31

DOMAIN 31 · OPERATIONAL SHEET

Reports, dashboards and consolidation

Management reporting offers member, activity, association and federation views with common indicator definitions. A total must cover the entire selected filter, not the visible page of a list. Figures distinguish cash flows, receivables, commitments, income and result. Consolidation eliminates identified internal movements and preserves the breakdown by entity. Charts have axes, units, period, legend and data table. The manual's real data are used as documentary references, not as an up-to-date statement of the service.

Manual reference: p. 52–59, 75, 79, 82. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
View	List, kanban, line chart, bars, matrix	Choose according to the management question, not to multiply charts.
Scope	Member, activity, association, federation	Check rights before aggregation.
Time	Transaction date, value date, accounting date	The user sees the report's time basis.
Measure	Cash, outstanding, unpaid, commitments, result	Avoid a single "balance" without a definition.
Comparison	Budget, previous period, cohort, scenario	Comparable scopes and currencies.
Distribution	Screen, PDF, CSV, accounting export	Same filter, same snapshot and same visibility.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behaviour
REP01	Every indicator has a formula, source, period, exclusions, refresh frequency and owner.
REP02	Consolidation never adds amounts in different currencies without explicit conversion.
REP03	An internal transfer does not inflate consolidated revenue.
REP04	Small populations and sensitive data may require masking to prevent re-identification.

B · Form and data-entry experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced with fictitious values. Final validation runs server-side.

Functional field	Type or content	Validation and input assistance
D31-F01 · Scope	List of authorised entities	Show entities excluded for lack of rights.
D31-F02 · Period	Start, end, date basis	Explicit bounds, time zone and inclusivity.
D31-F03 · Filters	Activity, member, type, status	Applied before totals and pagination.
D31-F04 · Indicators	Versioned catalogue	Definitions accessible from each card.
D31-F05 · Presentation currency	Code and rate policy	Source, date and rate method retained.
D31-F06 · Consolidation	Scope and eliminations	List of reciprocal flows and differences.
D31-F07 · Confidentiality	Export profile	Masking identical to that of the screen.
D31-F08 · Snapshot	Date, version, hash	Reproduce the report after later changes.

COMMON CONTEXT AND EDITING RIGHTS

The screen always shows the association, activity or file concerned, the currency where necessary, the status and the version. The author, creation date and identifier are generated by the system. The role authorised at the workflow step determines the available actions; a hidden button does not replace the access control.

> **Summary before validation.** The screen presents the inputs, the expected result and the errors before the final action. After submission, any significant correction displays the differences; errors target the field concerned and preserve the other entered data.

Screen actions: save a draft, verify, submit, request additional information, approve or reject according to role.

C · Workflow, exceptions and transactional control

Actor	Action	Control
Reader	Choose the scope	Rights checked
System	Freeze the data	Reference date
Engine	Compute the measures	Common definitions
Consolidator	Process reciprocal flows	Explicit differences
Controller	Validate	Reconciliation to the ledgers
System	Export	Filters and version included

EXCEPTIONS AND RECOVERY

Trigger	Response	Expected trace or recovery
Missing data	Show as incomplete	Do not automatically replace with zero.
Missing exchange rate	Block the converted total	Present subtotals by currency.
Intersection difference	Show a difference line	Do not force an unjustified elimination.
Large export	Run as a background job	Temporary link and rights re-checked at download.

Trigger	Response	Expected trace or recovery
Old report	Keep the snapshot	A new computation receives a new version.

D · Combinations, test recipes and indicators

COMBINATIONS AND QUALIFICATIONS

Combination of functions	Qualification	Validity condition
Consolidation + people with multiple memberships	Conditional	Distinguish unique persons from memberships.
Chart + sensitive data	Conditional	Aggregation and masking thresholds.
Budget + cash desk	Compatible	Separate columns for committed and paid.
Report + closing	Compatible	Signed or approved snapshot retained.
Scenario + actuals	Compatible	Distinct visual styles and visible assumptions.

TEST SCENARIOS

Case	Data or trigger	Verifiable result
D31-T01 · One hundred rows, page of twenty-five	Filter total	The total includes all one hundred rows.
D31-T02 · Internal transfer 100 000	Two consolidated sections	Consolidated external flow of zero.
D31-T03 · XAF and EUR without a rate	Request for a single total	Total refused; subtotals available.
D31-T04 · HORIZON history	101 579 832 minus 100 360 001	Reference balance 1 219 831.
D31-T05 · Member export	Filter over the whole association	Details limited to authorised rights.

MANAGEMENT INDICATORS

Indicator	Formula	Reading
Data freshness	Current instant minus last integrated event	Display the time on the dashboard.
Indicator traceability	Measures linked to source entries / financial measures	Navigation leads to the authorised detail.

D32

DOMAIN 32 · OPERATIONAL SHEET

Security, authorisations and data protection

The target combines roles, scopes, attributes, mandates and separation of duties. Authentication proves who is acting; authorisation determines on which association, which resource and which operation. Technical administrators do not receive an implicit business right over funds. Exceptional support access is temporary, justified and visible in the audit trail. Sensitive documents, exports, background jobs and caches apply the same rules as screens. Secrets and passwords never appear in logs or in diagnostic artefacts.

Manual reference: p. 13–17, 58–59, 67–70, 84. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Authentication	Password, second factor, controlled recovery	Strengthen privileged accounts and sensitive operations.
Authorisation	Role, association, fund, amount, mandate	Evaluate server-side on every access.
Separation	Creation, approval, payment, control	Configurable incompatibilities with a security minimum.
Support	Exceptional access with limited duration	Reason, approval, log and expiry.
Data	Public, internal, confidential, sensitive	Minimal collection and masking by purpose.
Audit	Actions, sensitive reads, exports, rights changes	Protected log and retention policy.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behaviour
SEC01	Every request verifies the relationship between user, association and requested object, including by direct identifier.
SEC02	A mandate revocation takes effect without waiting for a new long-lived session.
SEC03	Stored objects and exports are not reachable through a guessable public URL.
SEC04	Financial movements are corrected by linked operations and never deleted, including by an administrator.

B · Form and data-entry experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced with fictitious values. Final validation runs server-side.

Functional field	Type or content	Validation and input assistance
D32-F01 · User and scope	References	Association, fund, activity or mission clearly bounded.
D32-F02 · Role and permissions	Action matrix	Distinguish view, create, approve, pay, export.
D32-F03 · Mandate	Start and end	Automatic expiry and separate substitution.
D32-F04 · Action ceiling	Amount and currency	Cumulative control if requests are split.
D32-F05 · Conflicts of functions	List of rules	Show incompatibilities before assignment.
D32-F06 · Support access	Reason, duration, approver	No implicit unlimited delegation.
D32-F07 · Retention policy	Category and duration	Legal hold and verified purge.
D32-F08 · Security incident	Severity, facts, actions	Preserve evidence and minimise exposed data.

COMMON CONTEXT AND EDITING RIGHTS

The screen always shows the association, activity or file concerned, the currency where necessary, the status and the version. The author, creation date and identifier are generated by the system. The role authorised at the workflow step determines the available actions; a hidden button does not replace the access control.

> **Summary before validation.** The screen presents the inputs, the expected result and the errors before the final action. After submission, any significant correction displays the differences; errors target the field concerned and preserve the other entered data.

Screen actions: save a draft, verify, submit, request additional information, approve or reject according to role.

C · Workflow, exceptions and transactional control

Actor	Action	Control
Manager	Request access	Mission and duration
Authority	Approve	Least privilege
System	Activate	Mandate and control
User	Execute	Server-side authorisation
Auditor	Review	Actions and anomalies
System	Revoke	Expiry or decision

EXCEPTIONS AND RECOVERY

Trigger	Response	Expected trace or recovery
Compromised account	Revoke the sessions	Secure recovery and review recent actions.
Object from another association	Deny access	Do not reveal its content in the error message.
Export after revocation	Block the download	Re-check rights despite the file having been created earlier.
Log unavailable	Suspend sensitive operations if necessary	No financial transaction without a durable trace.

Trigger	Response	Expected trace or recovery
Personal data request	Verify the requester	Provide a limited export and handle retention by category.

D · Combinations, test recipes and indicators

COMBINATIONS AND QUALIFICATIONS

Combination of functions	Qualification	Validity condition
System admin + board	Conditional	Two explicitly separated scopes and decisions.
Support + health	Restricted	Specific minimal access, never global by default.
Multi-association + cache	Mandatory	Keys and filters include the authorisation scope.
Offline + revocation	Conditional	Limit local data and re-check every write.
Audit + confidentiality	Compatible	Log references and avoid copying sensitive documents.

TEST SCENARIOS

Case	Data or trigger	Verifiable result
D32-T01 · Loan ID from another association	Authenticated user	No read or modification possible.
D32-T02 · Treasurer requests their own payment	Separation enabled	A second approver is mandatory.
D32-T03 · Mandate expiring at noon	Action attempted at 12:01	Authorisation denied.
D32-T04 · Copied document link	User without rights	Download denied.
D32-T05 · Account reset	Audit log	No secret or password is recorded.

MANAGEMENT INDICATORS

Indicator	Formula	Reading
Active privileged accesses	Accounts with non-expired sensitive power	Periodic review by a named owner.
Isolation violations	Cross-association access tests or incidents	Acceptance target: zero unauthorised access.

D33

DOMAIN 33 · OPERATIONAL SHEET

Data architecture, API and integrations

The documented foundation combines Next.js, React, TypeScript, Prisma and PostgreSQL. The target favours a modular monolith with clear business boundaries first, rather than a premature multiplication of services. The transactional core contains memberships, contracts, obligations, payments, allocations and the ledger. Connectors and notifications run after transactional validation through a reliable queue. A domain event represents an established fact, distinct from a requested command. Technical proposals must be confronted with the actual schema and code before implementation; this study is not an audit of the repository.

Manual reference: p. 9–13, 69–70, 86. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Modularity	Organisation, finance, social, activities, resources	Explicit domain interfaces and owning invariants.
Persistence	Normalised relations, typed configurations	Database constraints complement application controls.
Transactions	Atomic write, reservation, locking	Prevent double spending and double allocation.
Asynchrony	Outbox, queue, recovery, error queue	Tolerate repetition and failure without losing the event.
API	Commands, queries, exports, webhooks	Versioned contracts and structured errors.
Interoperability	Payment, accounting, identity, messaging	Replaceable adapters with contract tests.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behaviour
ARC01	The transaction records the business state and the event to publish in the same durable commit.
ARC02	Consumers accept repeated delivery and make their effects idempotent.
ARC03	Unique constraints and scope keys prevent inconsistencies even under concurrency.
ARC04	Amounts, currencies and value dates are domain types, not free-form strings.

B · Form and data-entry experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced with fictitious values. Final validation runs server-side.

Functional field	Type or content	Validation and input assistance
D33-F01 · Command	Name and version	Proposed example: approveAid, confirmPayment, allocateTontine.
D33-F02 · Request identifier	UUID	Correlation between interface, API and log.
D33-F03 · Idempotency key	Bounded text	Associated with the body hash and the scope.
D33-F04 · Association context	Identifier	Checked against the session, never accepted on trust.
D33-F05 · Object version	Expected integer	Explicit conflict if the object changed since it was read.
D33-F06 · Business data	Validated schema	Exact amounts, references and controlled units.
D33-F07 · Outbound event	Type and minimal payload	No secret or medical document in the bus.
D33-F08 · Error	Code, reason, resolution	Distinguish validation, conflict, unavailability and access denial.

COMMON CONTEXT AND EDITING RIGHTS

The screen always shows the association, activity or file concerned, the currency where necessary, the status and the version. The author, creation date and identifier are generated by the system. The role authorised at the workflow step determines the available actions; a hidden button does not replace the access control.

> **Summary before validation.** The screen presents the inputs, the expected result and the errors before the final action. After submission, any significant correction displays the differences; errors target the field concerned and preserve the other entered data.

Screen actions: save a draft, verify, submit, request additional information, approve or reject according to role.

C · Workflow, exceptions and transactional control

Actor	Action	Control
Client	Send a command	Context and key
API	Authenticate and authorise	Scope verified
Domain	Validate the invariants	Version and limits
Database	Commit state and outbox	Atomicity
Worker	Publish and integrate	Idempotent recovery
Observability	Track the outcome	Full correlation

EXCEPTIONS AND RECOVERY

Trigger	Response	Expected trace or recovery
Failure after commit	Resume from the outbox	The business operation is not lost.
Failure before commit	Roll back the transaction	No partial state and no success notification.
Repeated message	Deduplicate by event	Single financial effect.
Unknown API version	Refuse clearly	Do not silently interpret a different structure.
Connector unavailable	Queue with a bounded wait	Error circuit and controlled recovery.

D · Combinations, test recipes and indicators

COMBINATIONS AND QUALIFICATIONS

Combination of functions	Qualification	Validity condition
Rule JSON + strong types	Compatible	Versioned schema and controlled migrations.
Financial ledger + events	Compatible	The ledger remains the source of amounts; events carry the facts.
Microservice + financial transaction	Conditional	To be introduced only with a demonstrated need and a consistency protocol.
Public API + private documents	Restricted	Temporary URLs and authorisation checks.
Webhook + audit	Compatible	Verified signature, minimised content, reference retained.

TEST SCENARIOS

Case	Data or trigger	Verifiable result
D33-T01 · Two concurrent withdrawals	Balance insufficient for both	At most one transaction consumes the available funds.
D33-T02 · Crash after write	Event not yet sent	The worker publishes it after restart.
D33-T03 · Same key, different body	Amount changed	Conflict returned, no second operation.
D33-T04 · Cross-association reference	Valid object in another scope	Application and relational constraints prevent the link.
D33-T05 · Long export	Interactive API	Background processing and status tracking.

MANAGEMENT INDICATORS

Indicator	Formula	Reading
Duplicated effects	Number of improper entries for the same command	Must remain zero in recovery and concurrency tests.
Business latency	Duration from command to commit	Measure p95 and p99 under representative load.

D34

DOMAIN 34 · OPERATIONAL SHEET

Mobile, accessibility, languages and notifications

The interface must remain usable on a phone, an unstable network and a shared device. The manual describes a responsive web app; offline, USSD and transactional SMS are extensions. Offline mode keeps drafts and a limited cache; it must not confirm an expense or a sensitive vote without central validation. Amounts always display currency and period. Translations also cover errors, receipts and notifications, with a visible fallback. Accessibility relies on keyboard navigation, contrast, labels, reading order and information independent of colour.

Manual reference: p. 60–64, 71–72, 82. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Channel	Web, mobile, PWA, future USSD or SMS	Explicit capabilities and limits per channel.
Connection	Online, cached read, draft, synchronisation	Visible state and no misleading local financial confirmation.
Language	French, English, other locales	Version and coverage tracked; business terms validated.
Notification	In-app, e-mail, SMS, WhatsApp	Consent, confidentiality and cost per channel.
Preference	Reminders, digest, times, priority channel	Necessary communications have a separate policy.
Accessibility	Keyboard, screen reader, zoom	Test financial journeys, not only the home page.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behaviour
MOB01	Synchronisation revalidates identity, mandate, version and availability before any write.
MOB02	A draft synchronised several times creates only one request thanks to its stable key.
MOB03	Notifications contain no diagnosis or sensitive details visible on a locked screen.
MOB04	A translation changes neither a numeric value, nor a currency, nor the meaning of a decision.

B · Form and data-entry experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced with fictitious values. Final validation runs server-side.

Functional field	Type or content	Validation and input assistance
D34-F01 · Language preferences	Locale code	Explicit fallback if a screen is not translated.
D34-F02 · Time zone and format	User settings	Economic dates remain attached to the contract.
D34-F03 · Contact channels	Verified contact details	Verification before sending sensitive messages.
D34-F04 · Consents	Choice per use	Distinguish service reminder from optional communication.
D34-F05 · Receiving hours	Windows and zone	Urgency according to the announced policy.
D34-F06 · Local draft	Identifier and version	Minimal storage, expiry and device protection.
D34-F07 · Synchronisation status	Status and errors	Show conflict, rejection or server success.
D34-F08 · Accessibility preferences	Size, contrast, help	Maintain labels and navigation order.

COMMON CONTEXT AND EDITING RIGHTS

The screen always shows the association, activity or file concerned, the currency where necessary, the status and the version. The author, creation date and identifier are generated by the system. The role authorised at the workflow step determines the available actions; a hidden button does not replace the access control.

> **Summary before validation.** The screen presents the inputs, the expected result and the errors before the final action. After submission, any significant correction displays the differences; errors target the field concerned and preserve the other entered data.

Screen actions: save a draft, verify, submit, request additional information, approve or reject according to role.

C · Workflow, exceptions and transactional control

Actor	Action	Control
User	Prepare the action	Legible amounts
Application	Save the draft	Clear offline state
Sync service	Send	Stable key
Server	Revalidate	Rights and version
Application	Show the result	Success or correction
Notification	Confirm soberly	Channel and preference

EXCEPTIONS AND RECOVERY

Trigger	Response	Expected trace or recovery
Connection lost after sending	Check the status	Do not immediately invite the user to pay again.
Outdated local version	Present the conflict	The user reviews the changed conditions.
Lost device	Revoke access	Limit the data kept locally.
SMS not delivered	Try the planned channel	Do not multiply sends without limit.
Missing translation	Show the fallback	Keep the message code for correction.

D · Combinations, test recipes and indicators

COMBINATIONS AND QUALIFICATIONS

Combination of functions	Qualification	Validity condition
Offline + cash	Conditional	Numbered provisional receipt then reconciliation, not a locally confirmed balance.
USSD + complex loan	Conditional	Consultation and simple request; detailed contract on a suitable channel.
Multilingual + rules	Compatible	Stable business identifiers and translated labels.
Notification + quorum	Distinct	A delivered message does not mean attendance or a vote.
Shared device + session	Conditional	Locking, sign-out and no persistent sensitive cache.

TEST SCENARIOS

Case	Data or trigger	Verifiable result
D34-T01 · Offline contribution draft	Network return	A single intention; confirmation only after server proof.
D34-T02 · Mandate revoked while disconnected	Attempted synchronised validation	Refusal with an understandable message.
D34-T03 · Amount 10 000 XAF	Language changed	Value and currency unchanged.
D34-T04 · Screen reader	Error on an amount	The field and the expected correction are announced.
D34-T05 · Medical reminder	Locked screen	Generic message without any health reason.

MANAGEMENT INDICATORS

Indicator	Formula	Reading
Synchronisation success	Confirmed drafts / submitted drafts	Distinguish business conflicts from network errors.
Journey accessibility	Critical journeys verified / defined journeys	Measure keyboard, screen reader and zoom.

D35

DOMAIN 35 · OPERATIONAL SHEET

Administration, migration, operations and support

The move to ARP must preserve existing data and historical evidence. Administration manages templates, features, quotas, countries, languages, support and monitoring without directly correcting business balances. Migration follows inventory, mapping, cleansing, dry run, reconciliation and a controlled cutover. A backup is not enough: a restore must be exercised and measured. Rolling back an application does not guarantee the compatibility of a data schema; migrations must be prepared for temporary coexistence of versions or a validated recovery plan.

Manual reference: p. 65–70, 82–87. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Migration	Initial, incremental, spreadsheet migration	Retain provenance, source identifiers and batch hash.
Activation	By association, module, pilot cohort	Blocking dependencies are checked.
Backup	Database, files, configurations, protected keys	Consistent restore and isolated environment.
Deployment	Staging, canary, cutover, rollback	Tests and migration compatibility.
Support	Incident, question, defect, change request	Explicit severity, owner and due date.
Monitoring	Availability, queues, errors, backups	Actionable alert with procedure and owner.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behaviour
OPS01	An import re-run with the same batch duplicates neither members nor movements.
OPS02	Migration totals reconcile by member, activity, account and currency before cutover.
OPS03	A restore is tested through to reading the documents and reconciling the balances.
OPS04	A production fix does not alter entries without a business procedure and an audit.

B · Form and data-entry experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced with fictitious values. Final validation runs server-side.

Functional field	Type or content	Validation and input assistance
D35-F01 · Import batch	Identifier and origin	Source file retained with restricted access.
D35-F02 · Mapping	Columns to fields	Units, dates, currencies, statuses and references.
D35-F03 · Anomalies	Rejected rows and reasons	No silent correction of an amount.
D35-F04 · Control totals	Accounts and subtotals	Before and after import, detailed differences.
D35-F05 · Cutover plan	Window and owners	Freeze, backup, final import and validation.
D35-F06 · Recovery objectives	Proposed RPO and RTO	Values to approve after tests, not unmeasured promises.
D35-F07 · Incident	Severity and impact	Number of associations, functions and funds affected.
D35-F08 · Restore plan	Steps and evidence	Isolated environment and consistency checks.

COMMON CONTEXT AND EDITING RIGHTS

The screen always shows the association, activity or file concerned, the currency where necessary, the status and the version. The author, creation date and identifier are generated by the system. The role authorised at the workflow step determines the available actions; a hidden button does not replace the access control.

> **Summary before validation.** The screen presents the inputs, the expected result and the errors before the final action. After submission, any significant correction displays the differences; errors target the field concerned and preserve the other entered data.

Screen actions: save a draft, verify, submit, request additional information, approve or reject according to role.

C · Workflow, exceptions and transactional control

Actor	Action	Control
Migration team	Inventory	Sources and quality
Business	Validate the mapping	Meaning of the data
Technical	Run a dry import	Idempotency
Controller	Reconcile	Totals and documents
Cutover owner	Authorise go-live	Criteria met
Operations	Monitor and restore	Tested procedure

EXCEPTIONS AND RECOVERY

Trigger	Response	Expected trace or recovery
Migration difference	Block the cutover	Identify the faulty rows and rules.
Duplicated source identifier	Quarantine	Documented manual resolution.
Incompatible migration	Stop the deployment	Data recovery plan, not a simple code rollback.
Unusable backup	Declare an incident	Restore a tested chain and review the risk.
Connector down	Enable degraded mode	Visible pending entries and later reconciliation.

D · Combinations, test recipes and indicators

COMBINATIONS AND QUALIFICATIONS

Combination of functions	Qualification	Validity condition
Template + existing association	Conditional	Proposed update, simulation and acceptance.
Quota + financial activity	Conditional	Do not block the repayment of an existing obligation.
Migration + anonymised data	Compatible	Staging without unnecessary personal data.
Support + balance correction	Restricted	Use an approved business correction operation.
Backup + encryption	Compatible	Also test the controlled availability of the keys.

TEST SCENARIOS

Case	Data or trigger	Verifiable result
D35-T01 · Import run twice	Same hash and references	No duplicated movement.
D35-T02 · HORIZON migrated	Principal 88 360 001 and interest 5 219 831	Subtotals found separately.
D35-T03 · Test restore	Database and files	Reports and supporting documents remain linked.
D35-T04 · Partial deployment	Old client still active	Planned compatibility or controlled maintenance.
D35-T05 · Incident resolved	Fix applied	Business validation and report before closure.

MANAGEMENT INDICATORS

Indicator	Formula	Reading
Migration difference	Target totals minus validated source totals	Zero tolerance on the amounts to migrate, except approved differences.

Indicator	Formula	Reading
Tested recovery time	End of checks minus start of restore	Compare with the approved objective and publish the date of the last exercise.

D36

DOMAIN 36 · OPERATIONAL SHEET

Internationalisation, currencies and country models

Internationalisation distinguishes interface language, entity country, contract currency, payment currency and consolidation currency. The manual announces several currencies, but also describes one currency per association and federal conversions to come. The target therefore retains a mandatory amount and currency on every contract, with one functional currency per entity. A country model gathers parameters and validation points, without claiming to constitute a legal certification. Savings, credit, fundraising and payment products are activated only after validation of the conditions applicable in the actual context.

Manual reference: p. 28, 71–72, 82. The requirements below describe the proposed target.

A · Scope, options and invariants

CONFIGURATION OPTIONS

Dimension	Options	Effect on the contract
Currency	Functional, contract, settlement, presentation	Do not assume equivalence of currency codes.
Conversion	Controlled entered rate, validated external source	Retain rate, date, source and any margin.
Difference	Exchange fee, gain or loss, rounding	Explicit breakdown in the ledger.
Country	Head office, branch, member, provider	These dimensions may be different.
Local model	Governance, data, accounting, payments	Version, validator and review date.
Activation	By country and by product	A global platform activation does not amount to local authorisation.

INVARIANTS TO MAKE VERIFIABLE

Rule	Mandatory behaviour
INT01	An amount without a currency is invalid; no direct addition of different currencies.
INT02	Currency precision comes from a controlled reference set, not from the number of decimals entered by the member.
INT03	The rate accepted on an operation stays fixed despite an update of the current rate.
INT04	Local models contain approved rules and activation reservations, without generalising the FEV or SLD examples.

B · Form and data-entry experience

The form below is a screen specification. Fields are required when their business condition applies; fields that do not apply are hidden, never replaced with fictitious values. Final validation runs server-side.

Functional field	Type or content	Validation and input assistance
D36-F01 · Entity country	Code and address	Distinguish declared head office from territory of activity.
D36-F02 · Bookkeeping currency	Code	Choice locked after the first entries unless a dedicated migration.
D36-F03 · Contract currency	Authorised code	Exposure and repayment policy.
D36-F04 · Payment amount	Value and currency	Show target amount, fees and net settled.
D36-F05 · Conversion rate	Exact decimal	Explicit rate direction, for example target units per source unit.
D36-F06 · Rate source and date	Reference	Supporting document or provider, timestamp and expiry.
D36-F07 · Country model	Version and validator	List of authorised products and remaining conditions.
D36-F08 · Compliance review	Decision and due date	Owner, scope and validation documents.

COMMON CONTEXT AND EDITING RIGHTS

The screen always shows the association, activity or file concerned, the currency where necessary, the status and the version. The author, creation date and identifier are generated by the system. The role authorised at the workflow step determines the available actions; a hidden button does not replace the access control.

> **Summary before validation.** The screen presents the inputs, the expected result and the errors before the final action. After submission, any significant correction displays the differences; errors target the field concerned and preserve the other entered data.

Screen actions: save a draft, verify, submit, request additional information, approve or reject according to role.

C · Workflow, exceptions and transactional control

Actor	Action	Control
Country owner	Define the context	Entities and products
Qualified experts	Validate the parameters	Framework and obligations
Product	Configure the model	Rules and limits
Finance	Test the conversions	Fees and rounding
Internal authority	Activate the scope	Documented decision
Controller	Review	Changes and incidents

EXCEPTIONS AND RECOVERY

Trigger	Response	Expected trace or recovery
Missing rate	Block the conversion	Keep the operation in its original currency.
Rate entered upside down	Detect through bounds and review	Show direction and example before confirmation.
Repayment in another currency	Apply the exchange clause	Compute the difference without changing the historical principal.
Product not validated in the country	Disable subscription	Keep consultation and settlement authorised.

Trigger	Response	Expected trace or recovery
Change of functional currency	Migration procedure	Snapshot and before/after reconciliation.

D · Combinations, test recipes and indicators

COMBINATIONS AND QUALIFICATIONS

Combination of functions	Qualification	Validity condition
Diaspora + local tontine	Conditional	Contract, currency and exchange risk accepted.
Federation + multi-currency	Compatible	Traced subtotals and presentation conversion.
English language + Cameroonian head office	Compatible	The language does not change the country of the contract.
Foreign payment + local aid	Conditional	Channel, identity and restrictions validated.
Country model + federal inheritance	Conditional	Resolve conflicts before publication.

TEST SCENARIOS

Case	Data or trigger	Verifiable result
D36-T01 · 100 source units, simulated rate 650	Target rate per source, no fees	65 000 target units; fictitious example, not a market rate.
D36-T02 · Rate updated after payment	Operation already confirmed	The historical receipt keeps its rate.
D36-T03 · XAF and XOF	Two distinct codes	No implicit merging of accounts.
D36-T04 · Conversion fee 500	Gross converted amount 65 000	Net 64 500 if fees are deducted from that amount.
D36-T05 · Expired country model	New sensitive subscription	Review required according to the activation policy.

MANAGEMENT INDICATORS

Indicator	Formula	Reading
Exposure by currency	Assets minus contractual liabilities by currency	Do not hide imbalances in the converted total.
Validated products	Products with an up-to-date review / products offered by country	Show activation restrictions.

APPENDIX 01 · UNITS AND CALCULATION RULES

Catalogue of financial formulas

All monetary variables carry a currency. Share quantities and rates are dimensionless; durations must be expressed in the unit of the rate. The examples do not set any universal price. Rounding is applied at the point defined in the contract, and the differences are then preserved.

Item	Reference formula	Conditions
Contribution	`C = q × u`	q = shares subscribed; u = price per period.
Net debt	`D = appel + ajustements − remises − affectations`	Present excess credit separately.
Distributable pot	`P = encaissements confirmés + report + financement − charges − réservations`	Include only the flows of the authorized fund.
Whole shares	`n = min(plafond, droits restants, plancher(P / b))`	b = amount per whole entitlement; then check eligibility.
Fractional shares	`q = min(plafond, droits, f × plancher(P / (b × f)))`	All limits are aligned on the f grid.
Constant-principal loan	`a = capital / n ; I(t) = CRD(t−1) × i`	i = rate per period; final principal adjustment.
Constant annuity	`A = capital × i / (1 − (1 + i)^(−n))`	For i = 0, A = capital / n.
Dated simple interest	`I = capital × taux annuel × jours / base`	Base explicitly chosen, for example 365.
Available savings	`S libre = S acquis − blocages − retraits réservés`	Eliminate overlaps between reservations.
Capped aid	`A = min(formule, reliquat événement, reliquat annuel, reliquat ménage)`	Approved but unpaid aid consumes the cap.
Dividend	`d(i) = base distribuable × poids(i) / somme des poids`	Base and liquidity checked separately.
Quorum	`Q = plafond supérieur(taux × base admissible)`	Recognized attendees ≥ Q; explicit base and proxies.

Largest remainder method: calculate the exact shares, allocate the whole units below, then distribute the remaining units to the largest remainders. In the event of a tie, apply a published stable order or another approved rule. The final sum must equal the base exactly.

APPENDIX 02 · CONTROLLED EXAMPLE OF 120 000 XAF OVER THREE MONTHS

Comparison of loan methods

Assumptions: principal disbursed immediately, rate of 3 % per month, three equal monthly instalments, no fees, no penalties. The constant-principal loan repays 40 000 of principal per month. The flat-rate method calculates interest each month on the initial principal.

Month	Opening principal outstanding	Constant principal	Declining interest	Total due
1	120 000	40 000	3 600	43 600
2	80 000	40 000	2 400	42 400
3	40 000	40 000	1 200	41 200
Total	—	120 000	7 200	127 200

Method	Calculation	Consequence
Constant principal	3 600 + 2 400 + 1 200 = 7 200	Instalments decrease with the outstanding principal.
Flat rate on initial principal	120 000 × 3 % × 3 = 10 800	Three instalments of 43 600; cost higher by 3 600.
Constant annuity	A = 120 000 × 0,03 / (1 − 1,03^(−3))	A ≈ 42 423,64 before rounding; last line adjusted.

Tests to automate: sum of principal = 120 000, final principal nil, non-negative interest for a positive rate, zero-rate branch and reproducibility after a language change. Early repayment must produce a new explicit state, without rewriting instalments already paid.

APPENDIX 03 · CONTRIBUTIONS AND PROMISED ENTITLEMENTS

Balance of a tontine contract

Twelve members, twelve sessions, a contribution of 50 000 and an award of 550 000 per session. The surplus is a resource to be allocated according to the contract, not automatically a personal profit for the treasurer.

Element	Per session	Full cycle
Contributions	12 × 50 000 = 600 000	7 200 000
Distributions	550 000	6 600 000
Surplus before other flows	50 000	600 000

Multiple shares and shortened duration

Four members with two shares and four with one share represent twelve shares. Two entitlements distributed per session exhaust the entitlements in six sessions, but that is not enough to prove the funding. With an illustrative price of 50 000 per share, each session collects 600 000. Two entitlements of 300 000 are fundable; two entitlements of 550 000 would create a deficit of 500 000 per session without an additional contribution.

> **Opening invariant.** Total planned contributions + identified funding ≥ promised distributions + planned charges + mandatory reserve. The system presents this equality to all subscribers before acceptance.

Fraction case: with b = 300 000, a minimum unit f = 0,25 and a pot of 250 000, three quarters of an entitlement cost 225 000, leaving 25 000. This distribution is permitted only if the contract allows beneficiary fractions and if the remaining entitlements allow them.

APPENDIX 04 · SENSITIVITY TO CONTRIBUTIONS AND EVENTS

Simulation of the solidarity fund

Illustrative simulation, distinct from the real SLD accounts: one hundred members, 6 000 XAF per month, twelve months, no initial cash. The collection rate is an assumption. The expenses constitute neither an actuarial forecast nor a promise of cover.

Scenario	Income	Simulated aid and costs	Annual balance
Central: 100 % collection	7 200 000	4 800 000	2 400 000
90 % collection	6 480 000	4 800 000	1 680 000
70 % collection	5 040 000	4 800 000	240 000
Shock: 90 % and three additional deaths	6 480 000	7 500 000	−1 020 000

The three additional deaths are simulated at 900 000 each, i.e. 2 700 000. The shock scenario requires funding or a formal decision on the uncovered obligations. A positive annual balance can hide a cash deficit during the year; a monthly schedule of aid and contributions must therefore complement this table.

> **ARP response.** Display the commitments already approved, the available reserve and the due dates. Prepare an exceptional call on an approved basis; never draw by default on refundable savings or another fund.

APPENDIX 05 · DOCUMENTARY DATA AND MIGRATION CONTROL

Reconciliation of the HORIZON case

The figures below are those of the anonymized manual, pp. 44, 54 and 57. They define the migration controls and do not represent a current consultation of the application.

Item	Amount XAF	Control
Capital paid in	8 000 000	Initial resource distinct from revenue.
Principal repaid	88 360 001	Reduces the loan receivables.
Interest received	5 219 831	Revenue identified in the history.
Total inflows	101 579 832	8 000 000 + 88 360 001 + 5 219 831.
Loans disbursed	100 360 001	Cumulative outflows; not current outstanding.
Net flow balance	1 219 831	Inflows less outflows.
Remaining principal	12 000 000	100 360 001 − 88 360 001.
Published dividend base	5 433 499	Distribution decision to be reconciled.
Gap, base less interest	213 668	Origin not explained in the elements cited.

Rounding of seven equal shares: 5 433 499 / 7 = 776 214,142857… XAF. To keep whole units, six members receive 776 214 and one member 776 215, according to an explicit tie-break. Displaying 776 214,14 for each share does not constitute a final distribution that preserves the sum.

> **Control before final import.** Reconcile the 49 loans, the 44 settled loans, the 5 active loans and the individual movements. Justify the dividend base by a decision and accounting entries. The portfolio/fund ratio of 150 % does not prove the availability of the cash required for payment.

APPENDIX 06 · COMPATIBILITY IS NOT AUTHORIZATION TO TRANSFER

Matrix of combinations between modules

A: standard coexistence with separate journals. C: conditional coexistence requiring an additional rule. The table is symmetric for coexistence; money flows remain directional and must be authorized separately.

Module	Tontine	Savings	Loan	Aid	Contrib.	Project	In-kind	Federation
Tontine	—	C	C	C	A	C	C	C
Savings	C	—	C	C	C	C	C	C
Loan	C	C	—	C	A	C	C	C
Aid	C	C	C	—	A	C	C	C
Contribution	A	C	A	A	—	A	C	A
Project	C	C	C	C	A	—	A	C
In-kind	C	C	C	C	C	A	—	C
Federation	C	C	C	C	A	C	C	—

Cross-cutting constraints to verify

Combination	Resolution rule
Tontine and loan	An advance to the beneficiary is a separate loan contract; it does not alter their tontine entitlements.
Savings and investment	A demand deposit does not automatically fund an illiquid asset; transformation limit and reserve.
Aid and debt	Aid remains a benefit; any set-off is explicit, authorized and visible to the beneficiary.
In-kind and cash distribution	The value of the good becomes liquidity only after sale or identified funding.
Federation and local funds	The funds remain separate; transfers are bilateral and eliminated only on consolidation.

Combination	Resolution rule
Payment and several obligations	A sum received may be split; the sum of the allocations never exceeds the payment.
Closing and active contract	Surviving receivables and entitlements are carried over explicitly; no erasure through a change of financial year.

Structural prohibitions. Double payment of the same order; two consumptions of the same entitlement; unauthorized cross-association access; deletion of a validated entry; addition of currencies without conversion; use of a restricted fund without authority.

APPENDIX 07 · REPRODUCIBLE METHOD AND COVERAGE LIMITS

Bounded combinatorial catalogue

The attached catalogue enumerates exactly 23 328 raw configurations across the ten axes below. Each row contains the chosen values and the conditions detected. This scope is finite and explicit: it does not claim to represent all possible amounts, schedules or future statuses.

Axis	Values covered
organisation	autonome, section, federation
adhesion	individuelle, familiale, institutionnelle
cotisation	fixe, parts, nature
tontine	aucune, rotation, tirage, encheres
credit	aucun, zero, amortissable
aide	aucune, forfait, jours
canal	especes, banque, mobile_money
devise	unique, conversion
connexion	en_ligne, hors_ligne
validation	simple, double

Test recipe coverage

A deterministic greedy algorithm selects 19 configurations covering the 351 distinct pairs of values between axes. The verified coverage is 100 % of the pairs of the raw domain. The conditions are not removed: they make it possible to test blocking or a parameterization request when a combination is not directly activatable.

The CSV files use the semicolon separator and UTF-8 encoding. The catalogue describes configurations, not operations actually executed. The test recipe file proposes the cases to be carried out on the product. Functional results are not declared successful by the mere generation of the files.

Methodological limit. Pairwise coverage does not prove three-dimensional interactions or calculation errors. It complements the 180 specific domain scenarios, the critical triples and the concurrency and recovery tests.

APPENDIX 08 · PRIORITY SCENARIOS BEYOND PAIRS

Critical three-dimensional interactions

Risk triple	Scenario to execute	Required result
Savings + guarantee + withdrawal	Two agents reserve a guarantee and then request a concurrent withdrawal.	Total committed never exceeds the balance; one action receives a conflict.
Tontine + fractions + auctions	Half an entitlement is subject to a discount expressed as a percentage.	Explicit basis for the half entitlement; quantity and amount preserved.
Aid + family + dual fund	Two spouses claim the same local and national aid.	Common event detected; cumulation allowed only per the rules of each fund.
Payment + network + repetition	Timeout, retry then two success webhooks.	One payment, one receipt and single allocations.
Loan + exit + guarantee	A beneficiary leaves the association before repayment.	Debt and guarantee retained; personal access adapted without loss of evidence.
Federation + foreign exchange + transfer	A section pays its contribution in another currency.	Two reciprocal entries, frozen rate and identified exchange difference.
Rule + retroactivity + closing	A price is changed after a session has been closed.	History unchanged; amendment or explicit adjustment in an open period.
Offline + revocation + validation	A revoked treasurer synchronizes an old payment draft.	Server-side rejection before any financial effect.
Dividend + variable capital + rounding	A member joins during the period and the base is not divisible.	Exact weight to date, final sum preserved and explicit tie-break.
Stock + in-kind + contribution	A contribution in sacks settles an obligation in cash.	Conversion and quality validated; stock increased, cash unchanged.

Each scenario retains the initial state, the commands, the order or concurrency, the responses, the final entries and the evidence. Add at minimum one forbidden-access case and one recovery-after-failure case for each financial journey. The final amounts must be recalculable independently of the interface.

APPENDIX 09 · ONE MEETING AND SEVERAL FUNDS

Integrated journey of an SLD-type section

Illustrative test scenario: twenty active members, one tontine share at 25 000, solidarity at 6 000, investment at 5 000 and refreshments at 3 000. The unit amounts come from the SLD model; the headcount of twenty members is a test assumption.

Step	Operation	Expected control
1 · Prepare	Freeze twenty memberships, rule versions and the session.	No call for a person excluded from the contract.
2 · Call	Per member: 25 000 + 6 000 + 5 000 + 3 000 = 39 000.	Expected total 780 000, split across four obligations.
3 · Collect	Nineteen pay 39 000; one pays 20 000.	Cash received 761 000 and residual debt 19 000.
4 · Allocate	The partial payment is allocated to the tontine per the authorized choice.	Tontine 495 000; solidarity 114 000; investment 95 000; refreshments 57 000.
5 · Award	Illustrative gross entitlement 450 000; 5 % discount on the entitlement, expressly chosen for this test.	Net 427 500; withholding 22 500; tontine balance 67 500 before allocation of the withholding.
6 · Pay refreshments	Flat amount 150 000, session income 57 000.	Funding deficit 93 000; payment blocked without carry-over or endowment.
7 · Review aid	Illness claim 70 000, entitlement verified.	Reservation against solidarity; fund available reduced to 44 000 before payment.
8 · Close	Reconcile accounts, allocations, entitlements and commitments.	No silent set-off between the four funds.

Discount control. In this example, 495 000 − 427 500 = 67 500 of remaining liquidity. This sum comprises 45 000 undistributed on the gross entitlement and 22 500 withheld. If the discount is transferred to another fund, two internal movements reduce the tontine balance to 45 000 without changing overall cash.

The example does not interpret the actual clause “5 % of the contribution rate”. It deliberately uses a named basis to demonstrate the expected behavior of the engine.

APPENDIX 10 · PRESERVING ENTITLEMENT TYPES

Integrated savings, loan and solidarity journey

A member holds 100 000 XAF of available savings. They request a loan of 120 000 over three months at 3 % monthly, constant amortization, with 60 000 pledged as collateral. The loan fund has the necessary liquidity and the partial guarantee is authorized by the product.

Step	Financial state	Evidence and control
Pledge	Savings held 100 000; blocked 60 000; available 40 000.	Consent and single reservation.
Disbursement	Loan 120 000; member debt 120 000 of principal.	Contract accepted and payment confirmed.
First instalment	Principal 40 000; interest 3 600; due 43 600.	Amortization method and version.
Partial payment of 20 000	Interest 3 600 then principal 16 400; principal outstanding 103 600.	Split and instalment remainder 23 600.
Separate aid of 70 000	After decision, social commitment of 70 000.	No implicit loan repayment by deducting the aid.
Withdrawal request of 50 000	Savings available 40 000 before other movements.	Refusal of 50 000; offer of the authorized amount or substitution of the guarantee.
Member departure	Loan, pledge and savings remain identified.	Liquidation statement with entitlements and debts separated.
Possible restructuring	New schedule, old arrears retained.	Member agreement and review of the guarantee.

Why avoid a single balance

A display that directly subtracts the loan from the savings and adds the aid mixes a receivable of the association, a debt to the member and a social benefit. The dashboard may present a summary, but must preserve these three positions and the restrictions on set-off.

Non-regression test. A correction of the 20 000 payment must reverse its allocations and its effects before re-posting. It must not modify the aid decision, the pledge contract or the loan method without a separate procedure.

APPENDIX 11 · PROPOSED ENTITIES AND CARDINALITIES

Target conceptual data model

Organization 1 → N Membership N → 1 Person

Membership carries category, dates, entitlements and local registration number.

Organization 1 → N Activity 1 → N Contract

The contract references a rule version and its participants.

Contract 1 → N Obligation; Payment N ↔ N Obligation

The Allocation table carries amount, currency and the related correction.

Transaction 1 → N Entry; Fund 1 → N Reservation

The journal keeps the movements; the reservation protects the available balance.

Complementary entity	Relations and constraints
Subscription / entitlement / award	Subscription linked to the member and the cycle; award consumes an entitlement, payment separate.
Loan / instalment / guarantee	A loan has schedule versions; a guarantee reservation links collateral and exposure.
Social event / claim / decision	One event may open several authorized entitlements; transactional committed caps.
Body / mandate / resolution	Mandate dated per person; resolution linked to a ballot and a document version.
Document / version / access	Immutable content, classification and validation evidence; temporary signed links.
Outbox / audit / idempotency	Event to be published atomically with state; action log; command uniqueness.

The arrows indicate conceptual cardinalities, not a Prisma schema extracted from the repository. Foreign keys must include or check the association scope; globally unique identifiers are not sufficient to prevent an incorrect cross-association link.

APPENDIX 12 · TECHNICAL RELIABILITY CONTRACT

Financial transaction and recovery after failure

1 · Client

Submit the command: stable key, amount, currency and expected version.

2 · API

Check the entitlements: session, association, mandate and separation of duties.

3 · Domain

Verify the invariants: available balance, cap, state and remaining entitlements.

4 · Database

Commit the transaction: state + journal + reservation + outbox event.

5 · Worker

Execute the external effects: idempotent connector, bounded retries.

6 · Reconciliation

Confirm the result: external evidence and handling of discrepancies.

Failure point	State to recover	Recovery action
Before commit to the database	No lasting effect.	Replay the command with the same key.
After commit, before response	Operation created but client uncertain.	Read the result by key; do not create a new operation.
After an operator call without response	External payment potentially made.	Query the status or reconcile before any new issuance.
After receiving a duplicate	Event already processed.	Respond with technical success without repeating the effect.
After a failed notification	Business operation successful.	Resume the notification only.

Proposed database constraints. Uniqueness of the provider reference; uniqueness of the command key per association and type; object version for concurrent update; non-zero amount constraint by type; debit/credit sums checked within the transaction; correction references to the original document. Business tables are not modified directly by connectors.

Message delivery may be repeated. It is the financial effects that must remain unique. The load test must include concurrency, failure and recovery, not only fast reads.

APPENDIX 13 · BATCHES AND VALIDATION GATES

Implementation trajectory

The sequence below proposes batches, without promising an unestimated schedule. The effort depends on the existing code, its test coverage, the data to be migrated, the providers and business trade-offs. Each batch can be deployed to a pilot association after acceptance testing.

Batch	Deliverable result	Exit gate
0 · Framing	Inventory of code and data, basis decisions, dictionary and priority models.	Blocking ambiguities resolved; scope signed off.
1 · Trust core	Dated memberships, authorizations, versioned rules, documents and audit.	Isolation and separation of duties tested.
2 · Shared finance	Obligations, payments, allocations, funds, accounts, reservations, journal.	Preservation of amounts and idempotent recovery demonstrated.
3 · Complete tontines	Subscriptions, entitlements, rotations, draws, auctions and closings.	All modes and priority defaults tested.
4 · Credit and savings	Products, schedules, guarantees, withdrawals, arrears and restructurings.	Formulas verified and concurrent caps safe.
5 · Solidarity and federation	Families, social entitlements, cumulations, split payments and consolidation.	Confidentiality, caps and reciprocal entries controlled.
6 · Integrations	Operator payments, notifications and accounting exports.	Provider contracts, sandbox and reconciliations validated.
7 · Extended resources	Projects, purchasing, stock, equipment, events and community.	Pilot journeys and control of earmarked funds.
8 · Geographic extension	Country models, languages, currencies and additional channels.	Local validation and operations able to support the scope.

Appoint a product owner for the trade-offs, an association lead for each profile, a finance lead for the formulas, an architect for the invariants, a quality lead for acceptance testing and an operations lead for recovery. Accountants, legal experts and providers intervene within their own scope before activation.

APPENDIX 14 · EVIDENCE EXPECTED BEFORE GO-LIVE

Acceptance plan and quality of service

Family	Proposed criterion	Acceptance evidence
Financial accuracy	Zero unexplained discrepancy on principal, payments, allocations and distributions.	Independent numeric test sets and reconciliations.
Isolation	No unauthorized access to another association or to a sensitive document.	Negative API, file, export and cache tests.
Concurrency	No double consumption of funds, entitlement, guarantee or stock.	Parallel tests and verification of the final state.
Recovery	A repeated command produces no additional financial effect.	Fault injection before and after commit.
Migration	Source and target totals equal after formally approved discrepancies.	Report per member, account, activity and currency.
Experience	Critical journeys achievable on a mobile screen and with the keyboard.	Usability and accessibility tests with acceptance data.
Performance	Initial target to be measured: p95 of current reads < 2 s outside the client network.	Representative load stating volume, concurrency and infrastructure.
Business continuity	Targets to be approved: RPO 24 h and RTO 4 h for a first pilot, to be strengthened according to risk.	Timed restoration of database and documents, final reconciliation.
Supervision	Payment, backup and queue incidents detected and assigned.	Alert exercises and procedures actually followed.
Documentation	Rules, forms and procedures agree with the delivered version.	Business review and update of the manual.

> **The objectives are not measured performance.** The latency, RPO and RTO values above are framing proposals. They must be accepted, adapted to the risks and demonstrated in the production environment. This study does not report the execution of tests on the application.

A blocking defect concerns in particular lost funds, double payment, wrong amount, data leak or absence of evidence. The go-live of a financial module waits for its correction and the repetition of the scenario concerned.

APPENDIX 15 · VOCABULARY SHARED BY TEAMS

Dictionary of management concepts

Concept	Operational definition
ARP	Association Resource Planning: integrated management of associative resources, commitments and activities.
Person and user	The person is a real actor; the user account allows them to sign in.
Membership	Dated relationship between a person and an association, carrying category, entitlements and obligations.
Activity and contract	The activity organizes a product; the contract freezes the conditions accepted by its participants.
Obligation	Amount or benefit due, with basis, due date, debtor and originating rule.
Payment	Flow received or issued, confirmed by evidence and attached to a channel.
Allocation	Monetary link between payment and obligation; allows partial and grouped payments.
Fund	Economic allocation of resources to a purpose; distinct from the bank account.
Reservation	Commitment of funds, entitlements or capacity; reduces the available balance without being a payment.
Beneficiary entitlement	Contractual quantity that can be awarded and then paid in a tontine.
Award	Decision designating a beneficiary; it does not prove the actual disbursement.
Principal outstanding	Capital remaining due on a loan, distinct from interest and penalties.
Pledge	Blocking of collateral as a guarantee under a validated contract and framework.
Aid and social loan	Definitive aid is not repayable; a social loan generates a schedule.
Dividend run	Versioned and approved calculation of a base to be distributed among beneficiaries.
Reversal	Reverse entry linked to an original entry, followed if needed by a new correct entry.
Idempotency	Same command repeated: same result achieved, without additional effect.

Concept	Operational definition
Outbox	Events to be published, stored atomically with the business transaction.
RPO and RTO	Maximum targeted data loss and targeted recovery time, to be validated by exercise.

The vocabulary shown to members may be simplified, but the data distinctions must remain stable. For example “contribute” may open a single interface that creates or settles an obligation, without conflating these operations in the model.

APPENDIX 16 · STUDY VALIDATION REGISTER

References and decisions to record

Primary source: Tchoua_Documentation_Complete_2026_Anonymisee.pdf, version 1.1, November 2026, 88 pages. The FEV and SLD texts, the HORIZON data and the repository specifications cited by this manual are not independently verified sources in this study. No market statistic or current legal rule has been added as an established fact.

Decision to record	Proposed owner	Expected effect
Auction basis and discount method	SLD model board and product	Unambiguous calculation accepted by subscribers.
Exact age, waiting period, compensated days and cumulations	Social committee	Complete catalogue with thresholds and cut-off dates.
Cumulation of sanctions and appeal route	Statutory body	Avoid double sanction and hidden interpretation.
Loan methods, rounding and displayed cost	Finance and product	Reproducible and comparable schedules.
Origin of the HORIZON dividend base	Treasurer and controller	Justification of the 213 668 XAF gap.
Nominal entitlements of the federation	Governance and data owner	Transparency compatible with confidentiality.
Authorized products and countries	Country leads and qualified experts	Activation limited to the validated scope.
Reserves, caps and set-off	GA or designated authority	Protection of funds and member understanding.
Recovery and retention targets	Operations and business owners	Measurable commitments and tested procedures.

Deliverables associated with this study

The PDF comprises 168 pages, 36 domains, 288 form-specific fields, 144 business invariants, 180 detailed scenarios and 180 commented combinations. The CSV catalogue contains 23 328 bounded configurations. The second CSV contains 19 cases covering all pairs of the ten declared axes.

Use in workshops. Assign each decision a status, an owner, a target date and approval evidence. Turn the domain rules and tests into backlog items. Validating a model prepares delivery; it does not replace software acceptance testing or the validation of applicable frameworks.